

# Planck-Scale Dimensional Reduction Without Topological Collapse

Spectral Dimension, Curvature Thresholds, and a Conditional TCGS-SEQUENTION  
Interpretation

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## Abstract

Planck units are fixed dimensional combinations of  $\hbar$ ,  $c$ , and  $G$ , but dimensional analysis alone does not determine the topological, Hausdorff, spectral, or operational dimension of the regime in which those units become relevant. The technical quantum-gravity literature contains a precise and nontrivial dimensional-reduction result: causal dynamical triangulations, asymptotic-safety constructions, Hořava–Lifshitz gravity, and several causal-set analyses obtain an effective or spectral dimension approaching approximately two in the ultraviolet. The relevant comparison is therefore between mathematically distinct notions of dimension rather than between Planck notation and an informal lower-dimensional slogan.

A separate issue concerns discreteness and Lorentz contraction. A universal value of  $\ell_P$  is not the coordinate edge of a rigid microscopic cell. Under an ordinary boost, a rest-frame cube of side  $\ell_P$  has longitudinal side  $\ell_P/\gamma_v$  and simultaneous three-volume  $\ell_P^3/\gamma_v$ . An observer-independent regular lattice of classical Planck cells would consequently select a preferred frame or require modified boost kinematics. This conclusion does not extend to discrete operator spectra, Lorentz-compatible causal sprinklings, scale-dependent spectral dimension, or invariant scalar thresholds.

For a positive diffusion generator  $\mathcal{D}$ , the return probability  $P(\sigma_D)$  defines

$$d_s(\sigma_D) = -2 \frac{d \log P(\sigma_D)}{d \log \sigma_D}.$$

A flow  $d_s \rightarrow 2$  is an analytic statement about propagation, heat-kernel scaling, the selected operator, and the probed ensemble. It can occur on a smooth manifold whose topological dimension remains four. Hořava’s anisotropic continuum construction supplies an explicit example. A topologically two-dimensional threshold is a different object: it can arise as the boundary of a three-dimensional admissible domain without any loss of carrier rank.

The differential-topological construction is formulated at a scale-free level. Let  $\mathcal{B}_C : \mathcal{C}^4 \rightarrow \mathbb{R}$  be a smooth dimensionless admissibility field, let  $\Omega_C = \{\mathcal{B}_C \leq 1\}$ , let  $\Gamma = \mathcal{B}_C^{-1}(1)$  be regular, and let  $X : \Sigma^3 \hookrightarrow \mathcal{C}^4$  be a rank-three immersion. Define

$$\mathcal{B}_\Sigma = \mathcal{B}_C \circ X, \quad \Omega_\Sigma = X^{-1}(\Omega_C), \quad \mathcal{H}_\Sigma = X^{-1}(\Gamma).$$

If  $X \pitchfork \Gamma$ , then  $\Omega_\Sigma$  is a smooth three-manifold with boundary  $\partial\Omega_\Sigma = \mathcal{H}_\Sigma$ ; the boundary may be empty, and every nonempty component of  $\mathcal{H}_\Sigma$  is two-dimensional. This is a standard preimage theorem and contains no intrinsic Planck physics. The notation  $\mathcal{B}_\Pi$ ,  $\Omega_\Pi$ , and  $\mathcal{H}_\Pi$  is warranted only after a separate normalization bridge links  $\mathcal{B}_C$  to  $\hbar$ ,  $c$ , and  $G$ . That bridge must either derive an intrinsic source normalization whose readout yields the observed constants or declare that the normalization is imported from a calibrated Shadow description. The second route is conditional on the unresolved source-to-Lorentzian-readout map and cannot be used to derive the constants it imports.

Selector covariance is likewise separated into two cases. Pure reparameterizations of one carrier make scalar naturality automatic and impose no restriction on  $\mathcal{B}_C$ . For a leaf-changing selector comparison with source lift  $\mathbf{f}_\kappa$  satisfying  $X_{\sigma'} \circ \varphi_\kappa = \mathbf{f}_\kappa \circ X_\sigma$ , naturality is equivalent to

$$X_\sigma^*(\mathbf{f}_\kappa^* \mathcal{B}_C - \mathcal{B}_C) = 0.$$

For a one-parameter lifted boost family generated by  $\xi$ , the corresponding local condition is  $\mathcal{L}_\xi \mathcal{B}_C = 0$  on the relevant orbit-saturated source region. Thus a covariant admissibility field must be a first integral of the comparison orbits. If  $\xi$  is transverse to a calibrated reference leaf and its flow sweeps a source region, the reference-leaf calibration and the first-integral condition form one characteristic initial-value problem: they determine the unique extension  $\mathcal{B}_C(\mathbf{f}_\eta(y)) = b_0(y)$  throughout the swept region. The physical derivation of the lifted comparison flow remains part of the causal-structure bridge.

A worked Schwarzschild calculation is kept distinct from the source preimage construction. On the Lorentzian Shadow spacetime  $(\mathcal{M}_{\text{Schw}}^4, g_{\text{Schw}})$ , the scalar

$$\hat{\mathcal{B}}_K = \ell_P^2 \sqrt{R_{\mu\nu\rho\sigma}[g_{\text{Schw}}] R^{\mu\nu\rho\sigma}[g_{\text{Schw}}]}$$

has unit level on the sphere

$$r_K = (2\sqrt{3})^{1/3} (\ell_P^2 r_s)^{1/3}.$$

For every Schwarzschild mass this is a two-sphere nested in a three-dimensional spatial regime; for macroscopic masses it lies deep inside the horizon. The special condition  $r_K = r_s$  gives  $M_\star = 0.9306 m_P$  and  $r_{s,\star} = 1.8612 \ell_P$ . The calculation is not an instance of the source-interface theorem unless a source-to-effective-metric bridge proves the required equality of fields. Gauss–Codazzi relations explain why pullback of a source curvature functional and construction of a curvature functional from induced data cannot be identified without proof.

Within TCGS–SEQUENTION, the interpretation is conditional. Minimal smooth completion is the parsimony premise that fixes four rather than five or more source dimensions, while exclusion of ontic time is the typing premise that classifies the additional direction as non-temporal. Neither premise follows from the preimage theorem. The immersion  $X$ , tensor pullback  $X^*$ , and typed 4D-Counterduction relation remain distinct. The source signature, the emergence of Lorentzian causal structure and invariant speed  $c$ , the source normalization package, the physical immersion, and the source-to-Schwarzschild bridge remain open. The  $\mathcal{A}_\Pi$  construction is retained only as a dimensionally consistent rescaling ansatz. The resulting claim is deliberately limited: ultraviolet spectral reduction, a topologically two-dimensional threshold, and a higher-dimensional carrier can coexist, but their compatibility does not by itself provide evidence for TCGS–SEQUENTION or any other source ontology.

**Keywords:** Planck scale; Planck-scale discreteness; Lorentz invariance; Lorentz contraction; preferred frame; spectral dimension; dimensional reduction; topological dimension; causal dynamical triangulations; Hořava–Lifshitz gravity; asymptotic safety; curvature invariants; transversality; preimage theorem; black-hole entropy; holography; TCGS–SEQUENTION; Counterspace; 4D-Counterduction.

# 1 Introduction

The Planck length, Planck time, Planck mass, Planck energy, and Planck density are conventionally written

$$\ell_{\text{P}} = \sqrt{\frac{\hbar G}{c^3}}, \quad t_{\text{P}} = \sqrt{\frac{\hbar G}{c^5}} = \frac{\ell_{\text{P}}}{c}, \quad m_{\text{P}} = \sqrt{\frac{\hbar c}{G}}, \quad (1)$$

$$E_{\text{P}} = m_{\text{P}} c^2 = \sqrt{\frac{\hbar c^5}{G}}, \quad \rho_{\text{P}} = \frac{m_{\text{P}}}{\ell_{\text{P}}^3} = \frac{c^5}{\hbar G^2}. \quad (2)$$

These expressions identify the dimensional intersection of quantum action, relativistic conversion, and gravitational coupling. They do not determine whether the microscopic regime is continuous, discrete, fractal, spectrally reduced, or topologically lower-dimensional. Meschini distinguished secure Planck-scale dimensional facts from stronger assumptions about the fundamental scale of a future spacetime theory, while Hossenfelder reviewed several independent mechanisms capable of producing minimum-length behavior [1, 2]. The ontological content of a Planck-scale proposal must therefore be supplied by additional structure.

A substantial technical literature studies ultraviolet dimensional reduction. Ambjørn, Jurkiewicz, and Loll measured a scale-dependent spectral dimension in causal dynamical triangulations, approaching four at large diffusion scales and approximately two in the ultraviolet [3]. Hořava obtained  $d_{\text{s}} = 1 + D/z$  for anisotropic scaling, giving  $d_{\text{s}} = 2$  for  $D = z = 3$  while retaining smooth four-dimensional topology [4]. Lauscher and Reuter obtained a microscopic spectral dimension of two in asymptotically safe Quantum Einstein Gravity, and Carlip surveyed related estimators across several quantum-gravity programmes [5–7]. These results motivate a precise distinction between spectral, topological, Hausdorff, boundary, and operational dimensions.

The analysis developed here has three layers. The first is framework-independent. It derives the Planck monomials, separates a universal constant from a rigid coordinate cell, reviews ultraviolet spectral flow, and shows why area normalization and black-hole entropy do not determine bulk topology. The second is differential-topological. It studies the preimage of a regular hypersurface under a rank-three immersion and proves that a nonempty transverse threshold is two-dimensional while its admissible domain remains three-dimensional. This theorem is scale-free and applies to any suitable four-manifold. The third layer is a conditional TCGS–SEQUENTION interpretation in which the source manifold, Shadow carrier, selector groupoid, and 4D-Counterduction architecture are assigned distinct mathematical types.

The source–Shadow notation requires particular care. The map

$$X : \Sigma^3 \hookrightarrow C^4$$

is an immersion or embedding selector, not a map from the source to the Shadow. Source tensors are pulled back by  $X^*$ . The constitutive dependence from a complete source configuration to an operational readout is represented separately by the typed 4D-Counterduction composition. Source regions and interfaces are related to the carrier by preimage. This separation follows the framework’s type-disciplined architecture and its distinction between source–Shadow realization and conventional holographic duality [8].

The conditional dimensional argument contains two independent load-bearing premises. A non-exhaustive rank-three carrier together with at least one independent normal relation establishes only  $\dim C \geq 4$ . The minimal-smooth-completion premise, explicitly a parsimony postulate, excludes

further independent source directions and fixes  $\dim C = 4$ . The exclusion of operational clock time and foliation labels from source dimensionality then classifies the added direction as non-temporal. Without minimality the source dimension may exceed four; without the no-ontic-time premise the four-dimensional completion may be ordinary  $(3 + 1)$  spacetime. Empirical anchors remain downstream constraints on particular maps rather than substitutes for either premise [9, 10].

Two further bridges are indispensable. First, the preimage theorem contains no  $\hbar$ ,  $c$ , or  $G$ . A source interface becomes a *Planck* interface only after a normalization map links the source admissibility field to the observed Planck package. Since the invariant speed  $c$  belongs to the causal structure of the effective Lorentzian Shadow description in the present architecture, importing  $\ell_P$  into a source field before deriving that readout would be a conditional calibration rather than a source-level derivation. Second, the source signature is not fixed by the boundary theorem. A Lorentzian source requires a mathematical discriminator separating Counterspace from ordinary spacetime, while a Riemannian source requires a derivation of the Shadow's Lorentzian metric, null cones, and invariant speed.

The readout-calibrated normalization and leaf-changing covariance are mathematically coupled rather than independent. Once a physically justified comparison flow is available, calibration on one transverse reference leaf supplies initial data for the transport equation  $\mathcal{L}_\xi \mathcal{B}_C = 0$ . The method of characteristics then gives a unique source-field extension on every region swept once by the comparison orbits. The unresolved burden is therefore concentrated in constructing the physical comparison action and its calibration data; the extension problem itself is well posed under explicit flow-box hypotheses.

Planck-scale discreteness introduces an independent covariance question. The universal constant  $\ell_P$  is not the coordinate edge of a classical cell. Ordinary Lorentz contraction is incompatible with a regular tessellation by rigid Planck cubes that is identical on every inertial simultaneity slice. That obstruction does not apply automatically to discrete operator spectra, Lorentz-compatible causal sprinklings, deformed relativistic kinematics, spectral flow, or scalar level sets. The relevant object and its transformation law must be specified before any covariance conclusion is drawn.

The  $G \mapsto G\mathcal{A}_\Pi$  construction is treated with the same discipline. It generates the exact rescaling

$$Q_\Pi = Q_P \mathcal{A}_\Pi^{(\ell+\tau-m)/2},$$

so products such as  $\ell_\Pi E_\Pi = \hbar c$  are algebraic cancellations rather than independent invariants. The ansatz is retained as a formal possibility but carries no physical burden until a geometric functional and a non-rescaling prediction are derived.

The paper proceeds as follows. Sections 2–4 establish the dimensional, source–Shadow, normalization, and transverse-preimage structure. Section 5 separates a universal Planck scale from a preferred microscopic lattice and derives the condition imposed on source admissibility fields by leaf-changing comparison maps. Section 6 analyses  $\ell_P^2$  curvature normalization. Section 7 treats the ultraviolet dimensional-reduction literature, and Section 8 separates area laws and dual descriptions from topological collapse. Section 10 supplies a complete Schwarzschild calibration while maintaining the source/Shadow distinction. Sections 9 and 14 state the unresolved bridges, formal scope, and decisive closure conditions.

## 2 What Standard Planck Units Establish

### 2.1 Dimensional construction

Let

$$Q = \hbar^a c^b G^d, \quad [Q] = M^m L^\ell T^\tau. \quad (3)$$

Using

$$[\hbar] = ML^2T^{-1}, \quad [c] = LT^{-1}, \quad [G] = M^{-1}L^3T^{-2}, \quad (4)$$

one obtains

$$a - d = m, \quad 2a + b + 3d = \ell, \quad -a - b - 2d = \tau. \quad (5)$$

The unique solution is

$$a = \frac{m + \ell + \tau}{2}, \quad b = \frac{m - 3\ell - 5\tau}{2}, \quad d = \frac{\ell + \tau - m}{2}, \quad (6)$$

and hence

$$Q_P(m, \ell, \tau) = \hbar^{\frac{m+\ell+\tau}{2}} c^{\frac{m-3\ell-5\tau}{2}} G^{\frac{\ell+\tau-m}{2}}. \quad (7)$$

Equation (7) is a dimensional theorem. Up to dimensionless numerical factors, the target dimensions fix the powers of  $\hbar$ ,  $c$ , and  $G$ . It does not specify the microscopic ontology of the regime in which the resulting quantity becomes relevant.

### 2.2 Four inferential levels

The phrase “Planck scale” should be separated into four claims.

- P1. Dimensional scale.** A quantity can be formed uniquely from  $\hbar$ ,  $c$ , and  $G$  for a prescribed dimensional target. This is closed by Eq. (7).
- P2. Operational pressure scale.** Quantum localization, relativistic propagation, and gravitational backreaction become jointly relevant near order-Planck combinations. This is motivated by effective-field-theory, localization, and collapse arguments [2, 11, 12].
- P3. Effective dimensional flow.** A specified probe, operator, ensemble, dispersion relation, or thermodynamic scaling law yields a scale-dependent dimension, for example  $d_s(\sigma_D) \rightarrow 2$ . This is a model-dependent but technically meaningful result, not a consequence of P1 alone.
- P4. Topological or ontological collapse.** The carrier itself loses a coordinate direction, changes topological dimension, or becomes an intrinsically two-dimensional substrate. This does not follow from P1–P3 without a separate rank, topology, dynamics, or observable-algebra theorem.

The distinction is especially visible in the density

$$\rho_P = \frac{m_P}{\ell_P^3}. \quad (8)$$

This is a volumetric mass scale in a three-dimensional spatial description. It does not prove that nature contains indivisible cubes of side  $\ell_P$ , but neither is it compatible with reading every Planck expression as inherently areal. The same formalism contains  $\ell_P^2$  when a curvature of weight  $L^{-2}$

is normalized and  $\ell_{\text{p}}^3$  when a volume is formed. The exponent follows the dimensional role of the quantity, not the topological dimension of the underlying domain.

### 2.3 Prior foundational caution

Meschini’s analysis is directly relevant because it identifies the often-unexamined step from the existence of natural units to their alleged inevitability in quantum gravity [1]. Hossenfelder’s review strengthens rather than weakens that caution: thought experiments, generalized uncertainty principles, string theory, loop quantization, noncommutative models, and asymptotic-safety arguments can motivate minimum-length behavior only through structures not contained in dimensional analysis alone [2]. The spectral-dimension literature adds a second epistemic ceiling. A derived ultraviolet flow can be physically informative while remaining an operator- and theory-relative statement. The correct non-entailments are

$$\text{Planck normalization} \not\Rightarrow d_{\text{s}} \rightarrow 2, \quad (9)$$

$$d_{\text{s}} \rightarrow 2 \not\Rightarrow d_{\text{top}} = 2, \quad \text{area scaling} \not\Rightarrow d_{\text{top}} = 2. \quad (10)$$

Model-derived spectral reduction is accepted within its stated operator and ensemble domains; only the unsupported promotion from effective dimension to carrier topology is excluded.

## 3 Typed Source–Shadow Architecture

### 3.1 The immersion is not the source-to-Shadow arrow

Let  $\mathcal{C}^4$  denote a smooth geometric representation of the source domain and let  $\Sigma^3$  denote a three-dimensional Shadow carrier. The admissible carrier map is

$$X : \Sigma^3 \hookrightarrow \mathcal{C}^4, \quad \text{rank}(dX) = 3 \quad (11)$$

on the regular sector. If  $G$  and  $\Psi$  denote source geometric and field data, the induced Shadow quantities are

$$(g, \psi) = (X^*G, X^*\Psi). \quad (12)$$

Equation (11) points from the carrier into the source because pullback is contravariant. Calling  $X$  a source-to-Shadow projection would reverse its mathematical direction. This paper therefore reserves the following terms:

$$\begin{aligned} X &: \text{immersion or carrier selector,} \\ X^* &: \text{pullback of declared source tensors or fields,} \\ \text{Ctd}_{4,S,G}^\sigma &: \text{constitutive source-to-Shadow readout relation.} \end{aligned} \quad (13)$$

The TCGS–SEQUENTIONS architecture types the last relation schematically as

$$\text{Ctd}_{4,S,G}^\sigma = \pi_\sigma^{\text{phys}} \circ \text{ev}_\sigma \circ \text{Read}_{S,G} \circ \text{Info}, \quad (14)$$

where a complete source configuration is mapped to its informational organization, then to a complete readability profile, then evaluated under selector  $\sigma$ , and finally registered as a physical observable [8]. Equation (14) is an order of constitutive dependence, not a temporal process and

not an additional manifold.

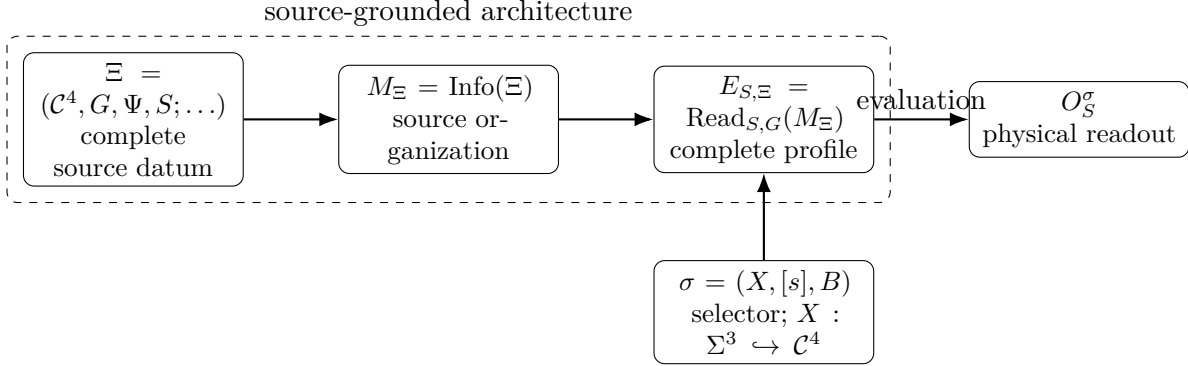


Figure 1: Typed architecture used in this paper. The immersion  $X$  is a selector appearing inside the constitutive readout architecture; it is not the arrow from source ontology to observed data. The arrows denote dependence, not temporal emission.

### 3.2 Why an embedding can still yield an incomplete Shadow description

An embedding preserves the intrinsic information carried by the embedded leaf together with its induced fields. It does not imply that the leaf determines the ambient source. The normal direction, inter-leaf relations, ambient curvature components, source fields outside  $X(\Sigma)$ , and distinctions lying in the null space of the forward readout can remain unidentifiable. In inverse-problem language, the map from source configurations to one readout can be many-to-one even when  $X$  itself is an embedding [13, 14]. The phrase “Shadow incompleteness” therefore means

$$(\Sigma, X^*G, X^*\Psi) \text{ need not determine } (\mathcal{C}^4, G, \Psi), \quad (15)$$

not that an embedding mysteriously deletes the points of its own image.

### 3.3 Conditional status of four-dimensional Counterspace

The formal source claim is conditional on three logically distinct premise classes; empirical diagrams and interference records do not by themselves supply the dimensional conclusion. Let  $\mathcal{P}_{\geq 4}$  collect the completion-need premises: a nontrivial family of rank-three registrations, source distinctions not exhausted by one registration, nonempty relative-invariant and artifact classes, and at least one independent normal or source relation not contained in the carrier tangent data. Let

$$\mathcal{P}_{\min} : \begin{array}{l} \text{the admissible smooth completion contains no additional} \\ \text{independent source directions beyond those required,} \end{array} \quad (16)$$

be the minimal-completion premise. Finally, let

$$\mathcal{P}_{-t} : \text{operational clock time and foliation labels are excluded from } \text{Dim}(C) \quad (17)$$

be the dimensional-typing premise. The inferential structure is then

$$\mathcal{P}_{\geq 4} \vdash \dim C \geq 4, \quad (18)$$

$$\mathcal{P}_{\geq 4} \wedge \mathcal{P}_{\min} \vdash \dim C = 4, \quad (19)$$

$$\mathcal{P}_{\geq 4} \wedge \mathcal{P}_{\min} \wedge \mathcal{P}_{\neg t} \vdash \text{the additional source direction is non-temporal.} \quad (20)$$

The exact count and dimensional type depend on two logically independent premises. The premise  $\mathcal{P}_{\min}$  fixes the count at four rather than five or more; it is a parsimony postulate about the smallest admissible smooth completion and is not derived from rank arithmetic alone. The premise  $\mathcal{P}_{\neg t}$  fixes the type of the added direction; without it, the same four-dimensional completion can be read as ordinary  $(3 + 1)$  spacetime. The remaining premises motivate source non-exhaustiveness and the need for at least one relation transverse to the carrier, but they do not independently settle either exact dimensionality or non-temporality.

The empirical anchors remain downstream. They can constrain a proposed source topology, metric, admissibility field, invariant estimator, or readout law; they do not replace  $\mathcal{P}_{\min}$  or  $\mathcal{P}_{\neg t}$ . This hierarchy follows the framework’s declared inference ceiling: empirical records pass through operational reconstruction and framework interpretation before any source-level claim is assessed [9, 10].

## 4 Regular Source Interfaces and Conditional Planck Normalization

### 4.1 Scale-free source field, admissible domain, and interface

Let

$$\mathcal{B}_C : \mathcal{C}^4 \longrightarrow \mathbb{R} \quad (21)$$

be a smooth dimensionless admissibility field. Define

$$\Omega_C := \{y \in \mathcal{C}^4 : \mathcal{B}_C(y) \leq 1\}, \quad \Gamma := \mathcal{B}_C^{-1}(1). \quad (22)$$

Assume  $d\mathcal{B}_C \neq 0$  on  $\Gamma$ . Then 1 is a regular value,  $\Gamma$  is a smooth hypersurface, and  $\Omega_C$  is a four-dimensional manifold with boundary

$$\partial\Omega_C = \Gamma, \quad \dim \Omega_C = 4, \quad \dim \Gamma = 3. \quad (23)$$

The adjective “dimensionless” means that a normalization has been selected. It does not yet identify that normalization as Planckian. The level-set geometry below depends only on smoothness and regularity.

### 4.2 Planck normalization as a separate bridge

**Definition 4.1** (Planck-normalization bridge). A source admissibility field  $\mathcal{B}_C$  is *Planck normalized* relative to a declared source–readout architecture only if there exist a source invariant package  $\mathcal{J}_C$ , a normalization rule  $\mathfrak{N}_C$ , and normalization data  $\mathfrak{P}$  such that

$$\mathcal{B}_C = \mathfrak{N}_C[\mathcal{J}_C; \mathfrak{P}], \quad (24)$$

and one of the following routes is established.

- (i) **Intrinsic-source route.** The source carries a normalization package  $\mathfrak{P}_C$  with a defined readout map  $\mathcal{R}_{\sigma_0}$  satisfying

$$\mathcal{R}_{\sigma_0}(\mathfrak{P}_C) = (\hbar, c, G)_{\sigma_0}, \quad (25)$$

and Eq. (24) uses  $\mathfrak{P} = \mathfrak{P}_C$  without importing the observed constants as prior source data.

- (ii) **Readout-calibrated route.** A reference selector  $\sigma_0$  supplies the measured constants  $(\hbar, c, G)_{\sigma_0}$  and a declared Shadow diagnostic. Let  $D_0 \subset \Sigma_{\sigma_0}$  be a calibration domain on which  $X_{\sigma_0}$  is an embedding, and define the embedded calibration leaf

$$\mathcal{L}_0 := X_{\sigma_0}(D_0). \quad (26)$$

For a curvature normalization, the calibration condition is

$$X_{\sigma_0}^* \mathcal{B}_C = \ell_{\mathbb{P}}^2 \mathcal{K}_{\sigma_0} \quad \text{on } D_0, \quad (27)$$

which determines smooth leaf data  $b_0 : \mathcal{L}_0 \rightarrow \mathbb{R}$  through

$$b_0 \circ X_{\sigma_0} = \ell_{\mathbb{P}}^2 \mathcal{K}_{\sigma_0} \quad \text{on } D_0. \quad (28)$$

Extension away from  $\mathcal{L}_0$  must then be supplied by selector-natural transport to physically equivalent readouts.

The two routes have different inferential status. Route (i) would constitute a source-level derivation of the normalization package. Route (ii) imports the Planck scale from the Shadow and therefore defines a calibrated cross-level diagnostic. It cannot be used to derive  $c$ ,  $G$ ,  $\hbar$ , or the Lorentzian readout structure on which the calibration depends. In the present TCGS–SEQUENTION architecture, where  $c$  belongs to the null structure of the effective Shadow metric, Route (ii) is conditional on the causal bridge in Eq. (86). Neither route is completed in this paper.

### 4.3 Carrier preimages

For the immersion  $X$  in Eq. (11), define

$$\mathcal{B}_\Sigma := X^* \mathcal{B}_C = \mathcal{B}_C \circ X, \quad (29)$$

and

$$\Omega_\Sigma := X^{-1}(\Omega_C), \quad \mathcal{H}_\Sigma := X^{-1}(\Gamma) = \mathcal{B}_\Sigma^{-1}(1). \quad (30)$$

Pullback acts on the scalar field; preimage acts on the source domain and interface. If Definition 4.1 is satisfied, the same objects may be denoted

$$\mathcal{B}_\Pi \equiv \mathcal{B}_\Sigma, \quad \Omega_\Pi \equiv \Omega_\Sigma, \quad \mathcal{H}_\Pi \equiv \mathcal{H}_\Sigma. \quad (31)$$

Without that bridge,  $\mathcal{H}_\Sigma$  is a regular admissibility threshold but not yet a physically derived Planck threshold.

Nothing in Eqs. (21)–(30) entails transversality. Transverse maps are generic in suitable function-space topologies, but a carrier selected by a constitutive admissibility rule is not an arbitrary

immersion. The realized crossing must be established by the source/readout construction.

**Theorem 4.2** (Transverse preimage theorem for a regular source interface). *Let  $\mathcal{C}^4$  and  $\Sigma^3$  be smooth manifolds, let  $\Omega_C = \{\mathcal{B}_C \leq 1\} \subset \mathcal{C}^4$  have regular boundary  $\Gamma = \mathcal{B}_C^{-1}(1)$ , and let  $X : \Sigma^3 \hookrightarrow \mathcal{C}^4$  be a rank-three immersion. If*

$$X \pitchfork \Gamma, \quad (32)$$

*then  $\Omega_\Sigma = X^{-1}(\Omega_C)$  is a possibly empty smooth codimension-zero submanifold of  $\Sigma^3$  with boundary*

$$\partial\Omega_\Sigma = \mathcal{H}_\Sigma = X^{-1}(\Gamma). \quad (33)$$

*Every nonempty connected component of  $\Omega_\Sigma$  has dimension three. Every nonempty connected component of  $\mathcal{H}_\Sigma$  is a smooth codimension-one submanifold of  $\Sigma^3$  and therefore has dimension two.*

*Proof.* The regular-value hypothesis makes  $\Omega_C$  an embedded codimension-zero submanifold of  $\mathcal{C}^4$  with boundary  $\Gamma$ . Transversality of  $X$  to  $\Gamma$  gives the preimage theorem for manifolds with boundary:

$$X^{-1}(\Omega_C) \text{ is a submanifold with } \partial X^{-1}(\Omega_C) = X^{-1}(\partial\Omega_C). \quad (34)$$

Because the domain  $\Sigma^3$  has dimension three, every nonempty component of the codimension-zero preimage has dimension three. If the boundary preimage is nonempty, its codimension in  $\Sigma^3$  is

$$\text{codim}_{\Sigma^3} X^{-1}(\Gamma) = \text{codim}_{\mathcal{C}^4} \Gamma = 1, \quad (35)$$

so its dimension is  $3 - 1 = 2$  [15].  $\square$

**Corollary 4.3** (Regularity of a nonempty pulled-back level). *Under the hypotheses of Theorem 4.2, if  $\mathcal{H}_\Sigma \neq \emptyset$ , then*

$$d\mathcal{B}_\Sigma \neq 0 \quad \text{on } \mathcal{H}_\Sigma. \quad (36)$$

*Proof.* At  $x \in \mathcal{H}_\Sigma$ , transversality means that  $dX_x(T_x\Sigma)$  is not contained in  $T_{X(x)}\Gamma = \ker(d\mathcal{B}_C)_{X(x)}$ . Hence  $(d\mathcal{B}_C)_{X(x)} \circ dX_x$  is nonzero. The chain rule identifies this composition with  $d(\mathcal{B}_C \circ X)_x = (d\mathcal{B}_\Sigma)_x$ .  $\square$

**Corollary 4.4** (Conditional Planck interpretation). *If Definition 4.1 is satisfied, the nonempty boundary  $\mathcal{H}_\Sigma$  may be denoted  $\mathcal{H}_\Pi$  and interpreted as a Planck-normalized threshold. Theorem 4.2 supplies its manifold structure and dimension; the Planck interpretation is supplied exclusively by the normalization bridge.*

**Corollary 4.5** (No regular dimensional collapse). *A nonempty regular threshold  $\mathcal{H}_\Sigma$  is two-dimensional while its admissible domain  $\Omega_\Sigma$  remains a three-dimensional manifold with boundary. The level condition does not imply*

$$\text{rank}(dX) < 3. \quad (37)$$

*A reduction of carrier dimension requires an independent rank-degeneracy or topology-change result.*

**Proposition 4.6** (Complementary intersection regimes). *Let  $\Gamma = \mathcal{B}_C^{-1}(1)$  be regular and let  $X : \Sigma^3 \hookrightarrow \mathcal{C}^4$  be a rank-three immersion.*

- (i) *If  $X(\Sigma) \cap \Gamma = \emptyset$ , then  $\mathcal{H}_\Sigma = \emptyset$ . On each connected component of  $\Sigma$ , continuity places the image entirely in  $\{\mathcal{B}_C < 1\}$  or entirely in  $\{\mathcal{B}_C > 1\}$ ; the corresponding component of  $\Omega_\Sigma$  is respectively all of that carrier component or empty.*

- (ii) If  $X(\Sigma) \subseteq \Gamma$ , then  $\mathcal{B}_\Sigma \equiv 1$ ,  $\mathcal{H}_\Sigma = \Sigma^3$ , and  $d\mathcal{B}_\Sigma = 0$ . The carrier is threshold-saturated and the threshold is three-dimensional.
- (iii) If  $X(\Sigma)$  meets  $\Gamma$  non-transversely without being contained in it,  $X^{-1}(\Gamma)$  may be singular, stratified, or of nonconstant dimension. Theorem 4.2 supplies no codimension formula in that regime.

**Criterion 4.7** (Carrier–interface crossing). A concrete realization may invoke the nonempty two-dimensional boundary conclusion only after it establishes  $\mathcal{H}_\Sigma \neq \emptyset$  and proves, for every  $x \in \mathcal{H}_\Sigma$ , the existence of  $v \in T_x \Sigma$  such that

$$(d\mathcal{B}_C)_{X(x)}(dX_x v) \neq 0. \quad (38)$$

Equivalently, the selected carrier must cross rather than follow the source interface.

*Remark 4.8* (Scale and evidentiary scope). Theorem 4.2 is standard differential topology. Its hypotheses contain no  $\hbar$ ,  $c$ , or  $G$ , and it applies to any smooth four-manifold, regular hypersurface, and transverse rank-three immersion, including ordinary spacetime with a spatial slice. It establishes compatibility between a three-dimensional carrier and a nonempty two-dimensional boundary. It neither identifies the boundary as Planckian nor distinguishes TCGS–SEQUENTION from other source interpretations.

*Remark 4.9* (Spectral scope). The theorem is silent about spectral dimension. A smooth three-dimensional domain can have a two-dimensional boundary while the ordinary bulk heat kernel has  $d_s = 3$ . Conversely, a topologically four-dimensional smooth spacetime can have  $d_s \rightarrow 2$  under an anisotropic or scale-dependent diffusion operator. Section 7 treats this independence.

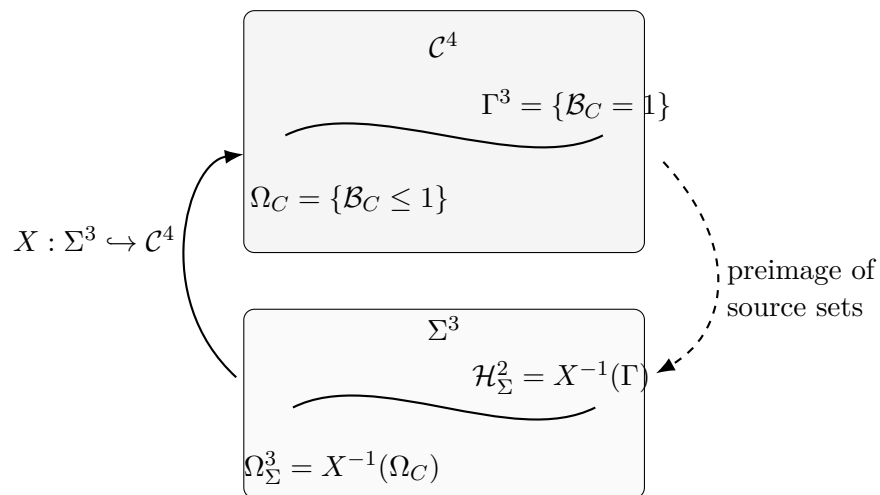


Figure 2: Source-interface and carrier-preimage hierarchy. The immersion points from the three-dimensional carrier into the four-dimensional source representation. A regular source hypersurface has a two-dimensional nonempty transverse preimage; the associated carrier domain is a three-manifold with boundary. Diagrammatic rectangles do not represent intrinsic dimension.

## 5 Planck-Scale Discreteness, Lorentz Contraction, and Covariant Thresholds

### 5.1 A universal constant is not a rigid coordinate cell

In a fixed unit convention, the Planck length

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (39)$$

is a universal dimensional constant. It is not transformed as though it were the coordinate length of an extended body. By contrast, the length assigned to an object is obtained by intersecting its world tube with an observer's simultaneity hypersurface and is therefore frame dependent. The two statements have different types:

$$\ell'_P = \ell_P, \quad L'_\parallel = \frac{L_0}{\gamma_v}, \quad \gamma_v = \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (40)$$

The first equation concerns a constant entering the laws; the second concerns an observer-dependent spatial interval of an extended rest-frame configuration.

**Proposition 5.1** (Preferred-lattice obstruction). *Assume a regular cubical tessellation of a spatial hypersurface by classical cells, each having rest-frame edge  $L_0 = \ell_P$  and rest volume  $V_0 = \ell_P^3$ . If ordinary Lorentz transformations hold, the same cells cannot have coordinate edge  $\ell_P$  and coordinate volume  $\ell_P^3$  on the simultaneity hypersurfaces of every inertial observer. A fundamental regular lattice of this kind therefore selects a preferred rest frame unless the boost law or the physical meaning of the cells is changed.*

*Proof.* For a boost of speed  $v$  parallel to one edge,

$$L'_x = \frac{\ell_P}{\gamma_v}, \quad L'_y = L'_z = \ell_P, \quad (41)$$

so the simultaneous three-volume in the boosted frame is

$$V' = L'_x L'_y L'_z = \frac{\ell_P^3}{\gamma_v}. \quad (42)$$

For  $v \neq 0$ ,  $V' \neq V_0$  and the cell is not a cube on the boosted simultaneity slice. Requiring the original cubical lattice to be physically privileged therefore identifies its rest frame. Requiring every inertial observer to recover identical cubical cells instead contradicts the undeformed Lorentz transformation used in Eq. (41).  $\square$

A single material object whose proper rest length happened to equal  $\ell_P$  would not by itself contradict relativity; it would behave like any other extended object and possess a rest frame. The inconsistency appears only when that rest frame is promoted to a universal microscopic lattice frame, or when  $\ell_P$  is required to be the same minimum *coordinate interval* for all inertial observers under ordinary boosts. Thus

$$L'_\parallel < \ell_P \not\Rightarrow \text{the constant } \ell_P \text{ has Lorentz-contracted.} \quad (43)$$

## 5.2 Discreteness is not one mathematical type

Proposition 5.1 is a no-go statement for a classical, regular, observer-independent lattice under undeformed Lorentz kinematics. It is not a no-go theorem for every form of Planck-scale discreteness. The technical literature supplies several distinct counterexamples to that overgeneralization.

In loop quantum gravity, the relevant discreteness is a spectrum of geometric operators rather than a tessellation by classical rods. Rovelli and Speziale showed that a minimum area eigenvalue need not be Lorentz-contracted as a fixed classical area: the spectrum can remain unchanged while a boost changes the probability distribution of measurement outcomes [16]. This result does not establish loop quantum gravity as the correct theory; it establishes that a discrete observable spectrum and a rigid lattice are categorically different.

Causal-set theory provides another distinction. A Poisson sprinkling into Minkowski spacetime is discrete without furnishing a measurable preferred spacetime direction; Dowker, Henson, and Sorkin developed Lorentz-invariant phenomenology for such discreteness, while Bombelli, Henson, and Sorkin proved that no equivariant measurable map from a sprinkling to a preferred direction exists [17, 18]. The price is nonlocal combinatorial structure rather than a finite-valency spatial lattice. Deformed-special-relativity constructions supply a third logical possibility by modifying the action of boosts so that a Planckian scale can be observer independent without a preferred inertial class [19]. Spectral dimensional flow supplies a fourth:  $d_s \rightarrow 2$  is a short-scale propagation law and need not posit discrete cells at all.

The correct hierarchy is therefore

$$\begin{aligned}
&\text{regular rigid lattice} \implies \begin{array}{l} \text{preferred frame, broken symmetry,} \\ \text{or deformed boosts,} \end{array} \\
&\text{discrete operator spectrum} \not\Rightarrow \text{regular spatial tessellation,} \\
&\text{Lorentz-compatible causal discreteness} \not\Rightarrow \text{preferred direction,} \\
&\hspace{15em} d_s \rightarrow 2 \not\Rightarrow \begin{array}{l} \text{lattice ontology} \\ \text{or topological collapse.} \end{array} \tag{44}
\end{aligned}$$

The word “discrete” is therefore insufficient unless the discrete object, its symmetry action, its continuum relation, and its observable meaning are declared.

## 5.3 The TCGS–SEQUENTION position: scalar admissibility rather than Planck cells

The TCGS–SEQUENTION architecture does not posit that  $\Sigma^3$  is tessellated by rigid cubes of edge  $\ell_P$ . Lorentz covariance, invariant intervals, light-cone relations, proper-time comparisons, and clock metrology are retained as Shadow-level structures, while the Lorentzian representation is not identified with the complete source ontology [8]. The framework’s discreteness analysis likewise separates finite records, lattice bases, discrete groups, and representation truncations [20].

When the Planck-normalization bridge of Definition 4.1 is available, the proposed Planck object is the scalar level set

$$\mathcal{H}_{\Pi}^{(\sigma)} = \left( \mathcal{B}_{\Pi}^{(\sigma)} \right)^{-1} (1), \tag{45}$$

not a coordinate cell of volume  $\ell_P^3$ . A coordinate transformation can alter coordinates assigned to a

level set without changing the scalar condition that defines it. In particular,

$$L'_{\parallel} < \ell_P \not\Rightarrow \mathcal{B}_{\Pi} > 1. \quad (46)$$

The Schwarzschild scalar  $\hat{\mathcal{B}}_K$  in Section 10 is an independent Shadow-spacetime diagnostic and is not identified with  $\mathcal{B}_{\Pi} = X^* \mathcal{B}_C$ .

**Definition 5.2** (Threshold-independent selector equivalence). For a fixed admissible source configuration  $\Xi$ , let  $\text{Sel}_{\Xi}$  be a groupoid whose objects are selectors  $\sigma = (X, [s], B)$ . An arrow  $\kappa : \sigma \rightarrow \sigma'$  exists only when threshold-independent data identify the selectors as physically equivalent presentations of one operational readout corridor. Such data include the effective metric or causal structure, calibrated rods and clocks, matter and detector readouts, and transported boundary or corridor information. Orientation-preserving relabelings of one foliation are canonical arrows. A change of immersion is an arrow only after an independent carrier-comparison map has been constructed.

For an arrow  $\kappa$ , let

$$\varphi_{\kappa} : \Sigma_{\sigma} \longrightarrow \Sigma_{\sigma'} \quad (47)$$

be its carrier comparison diffeomorphism. The scalar naturality condition is

$$\mathcal{B}_{\Sigma}^{(\sigma')} \circ \varphi_{\kappa} = \mathcal{B}_{\Sigma}^{(\sigma)}. \quad (48)$$

For a Planck-normalized field,  $\mathcal{B}_{\Sigma}$  may be replaced by  $\mathcal{B}_{\Pi}$ .

**Lemma 5.3** (Pure relabeling covariance). *Suppose  $\kappa$  changes only the presentation of one carrier and satisfies*

$$X_{\sigma'} \circ \varphi_{\kappa} = X_{\sigma}. \quad (49)$$

*Then Eq. (48) holds for every source scalar  $\mathcal{B}_C$ .*

*Proof.* Using  $\mathcal{B}_{\Sigma}^{(\sigma)} = \mathcal{B}_C \circ X_{\sigma}$ ,

$$\mathcal{B}_{\Sigma}^{(\sigma')} \circ \varphi_{\kappa} = \mathcal{B}_C \circ X_{\sigma'} \circ \varphi_{\kappa} = \mathcal{B}_C \circ X_{\sigma} = \mathcal{B}_{\Sigma}^{(\sigma)}. \quad (50)$$

□

Lemma 5.3 shows that reparameterization covariance is a consistency identity. It becomes a restrictive condition only when the comparison changes the embedded leaf.

**Definition 5.4** (Leaf-changing source lift). A leaf-changing selector arrow  $\kappa : \sigma \rightarrow \sigma'$  admits a source lift when there is a diffeomorphism  $\mathfrak{f}_{\kappa}$  between neighborhoods of the two carrier images such that

$$X_{\sigma'} \circ \varphi_{\kappa} = \mathfrak{f}_{\kappa} \circ X_{\sigma}. \quad (51)$$

**Proposition 5.5** (Admissibility-field criterion for leaf-changing naturality). *Let  $\kappa$  admit the source lift in Eq. (51). Then*

$$\mathcal{B}_{\Sigma}^{(\sigma')} \circ \varphi_{\kappa} - \mathcal{B}_{\Sigma}^{(\sigma)} = X_{\sigma}^* (\mathfrak{f}_{\kappa}^* \mathcal{B}_C - \mathcal{B}_C). \quad (52)$$

*Consequently, Eq. (48) holds if and only if the right-hand side of Eq. (52) vanishes. If a family of carrier images is orbit-saturating in a source region  $U$ , naturality for every lifted comparison in that family is equivalent to*

$$\mathfrak{f}_{\kappa}^* \mathcal{B}_C = \mathcal{B}_C \quad \text{on } U. \quad (53)$$

For a smooth one-parameter family  $\mathfrak{f}_\eta$  generated by a vector field  $\xi$ , the local condition is

$$\mathcal{L}_\xi \mathcal{B}_C = 0 \quad \text{on } U. \quad (54)$$

Thus admissible natural fields are first integrals of the comparison orbits and, when the quotient is regular, descend to  $U/\sim$ .

*Proof.* Equation (51) gives

$$\begin{aligned} \mathcal{B}_\Sigma^{(\sigma')} \circ \varphi_\kappa &= \mathcal{B}_C \circ X_{\sigma'} \circ \varphi_\kappa \\ &= \mathcal{B}_C \circ \mathfrak{f}_\kappa \circ X_\sigma = X_\sigma^*(\mathfrak{f}_\kappa^* \mathcal{B}_C). \end{aligned} \quad (55)$$

Subtracting  $X_\sigma^* \mathcal{B}_C$  yields Eq. (52). If the selected carrier images jointly cover the relevant comparison orbits, vanishing of all pullbacks is equivalent to Eq. (53) on the orbit-saturated region. Differentiation of  $\mathfrak{f}_\eta^* \mathcal{B}_C = \mathcal{B}_C$  at  $\eta = 0$  gives Eq. (54); conversely, vanishing Lie derivative integrates to invariance along the local flow of  $\xi$ .  $\square$

**Proposition 5.6** (Characteristic extension from a calibrated leaf). *Let  $\mathcal{L}_0 \subset U \subset \mathcal{C}^4$  be the embedded calibration leaf in Eq. (26), let  $b_0 \in C^\infty(\mathcal{L}_0)$  be the readout-calibrated data in Eq. (28), and let  $\xi$  be a smooth vector field on  $U$  that is transverse to  $\mathcal{L}_0$ . Let  $\mathfrak{f}_\eta$  denote its local flow. Assume that, for an interval  $I$  containing 0, the flow map*

$$\Phi : I \times \mathcal{L}_0 \longrightarrow U_{\text{sw}}, \quad \Phi(\eta, y) = \mathfrak{f}_\eta(y), \quad (56)$$

*is a diffeomorphism onto a swept source region  $U_{\text{sw}} \subset U$ . Then there exists a unique smooth admissibility field  $\mathcal{B}_C : U_{\text{sw}} \rightarrow \mathbb{R}$  satisfying*

$$\mathcal{B}_C|_{\mathcal{L}_0} = b_0, \quad \mathcal{L}_\xi \mathcal{B}_C = 0. \quad (57)$$

*It is given explicitly by*

$$\boxed{\mathcal{B}_C(\mathfrak{f}_\eta(y)) = b_0(y)}. \quad (58)$$

*Hence  $\mathfrak{f}_\eta^* \mathcal{B}_C = \mathcal{B}_C$  wherever both sides are defined. If 1 is a regular value of  $b_0$ , then 1 is a regular value of the extended field on  $U_{\text{sw}}$ , and*

$$\mathcal{B}_C^{-1}(1) \cap U_{\text{sw}} = \Phi(I \times b_0^{-1}(1)). \quad (59)$$

*Since  $\dim \mathcal{L}_0 = 3$ , regularity of the unit level gives  $\dim b_0^{-1}(1) = 2$ , while its characteristic saturation in Eq. (59) is a three-dimensional source hypersurface.*

*Proof.* Because  $\Phi$  is a diffeomorphism, define

$$\mathcal{B}_C := b_0 \circ \text{pr}_{\mathcal{L}_0} \circ \Phi^{-1}. \quad (60)$$

The field is smooth, equals  $b_0$  at  $\eta = 0$ , and is independent of the flow coordinate, so  $\mathcal{L}_\xi \mathcal{B}_C = 0$ . Every characteristic in  $U_{\text{sw}}$  intersects  $\mathcal{L}_0$  exactly once; therefore any other solution of Eq. (57) must take the same value on that characteristic, proving uniqueness. Flow invariance follows from Eq. (58). In the flow-box coordinates,  $d\mathcal{B}_C$  is the pullback of  $db_0$  in the leaf directions, so regularity of the unit level and Eq. (59) follow directly.  $\square$

Proposition 5.6 couples the readout-calibrated route of Definition 4.1 to the leaf-changing naturality condition of Proposition 5.5. Calibration fixes the field on one leaf, while source-orbit invariance propagates those data by characteristics. The remaining open inputs are physical rather than PDE-theoretic: the comparison flow must be derived independently, its generator must be transverse to the calibration leaf, and its orbits must sweep the stated region without multiple intersections or uncovered sectors.

**Corollary 5.7** (Covariant transport of a regular threshold). *If  $\kappa$  is an independently established arrow of  $\text{Sel}_\Xi$ , Eq. (48) holds, and  $\varphi_\kappa$  is a diffeomorphism, then*

$$\varphi_\kappa(\mathcal{H}_\Sigma^{(\sigma)}) = \mathcal{H}_\Sigma^{(\sigma')}, \quad \dim \mathcal{H}_\Sigma^{(\sigma)} = \dim \mathcal{H}_\Sigma^{(\sigma')} \quad (61)$$

whenever the thresholds are nonempty. Under Definition 4.1, the same statement holds for  $\mathcal{H}_\Pi^{(\sigma)}$ .

*Proof.* If  $x \in \mathcal{H}_\Sigma^{(\sigma)}$ , then  $\mathcal{B}_\Sigma^{(\sigma)}(x) = 1$ . Naturality gives  $\mathcal{B}_\Sigma^{(\sigma')}(\varphi_\kappa(x)) = 1$ . Applying the same argument to the inverse arrow gives equality of the level sets, and diffeomorphisms preserve dimension.  $\square$

The distinction between Lemma 5.3 and Proposition 5.5 is essential. Pure relabeling imposes no condition on  $\mathcal{B}_C$  beyond scalar composition. Leaf-changing covariance requires invariance along a source comparison action. In a Lorentzian effective theory, the construction of a lifted boost family  $\mathfrak{f}_\eta$  is part of the source-to-causality problem; once such a lift is available, Eq. (54) gives a determinate classification criterion for admissibility fields.

The framework position is therefore

$$\begin{aligned} \text{TCGS Planck scale} &= \text{conditionally normalized scalar admissibility threshold,} \\ \text{TCGS Planck scale} &\neq \text{rigid Planck cell.} \end{aligned} \quad (62)$$

Physical closure requires the normalization bridge, the independently constructed selector groupoid, the source lift of leaf-changing observer maps, and invariance of  $\mathcal{B}_C$  under the corresponding orbit action.

## 6 Why $\ell_P^2$ Does Not Mean Two-Dimensional Geometry

### 6.1 Curvature weight is not manifold dimension

Suppose  $\mathcal{K}_\Pi$  is a scalar curvature magnitude with

$$[\mathcal{K}_\Pi] = L^{-2}. \quad (63)$$

Define the purely dimensional normalization operator

$$\mathcal{N}_P[\mathcal{K}_\Pi] := \ell_P^2 \mathcal{K}_\Pi. \quad (64)$$

Then  $\mathcal{N}_P[\mathcal{K}_\Pi]$  is dimensionless. This symbol is deliberately not  $\mathcal{B}_\Pi$ : identification with the source-pulled scalar  $X^*\mathcal{B}_C$  requires one of the routes in Definition 4.1. Under the readout-calibrated route, Eq. (27) supplies the reference-leaf data and Proposition 5.6 supplies their unique off-leaf extension

only when its flow-box hypotheses hold. This construction is available on a two-dimensional surface, a three-dimensional Riemannian carrier, a four-dimensional Lorentzian spacetime, or a source manifold of other dimension. The physical dimension of curvature is fixed by two derivatives of the metric, not by the number of independent coordinates.

Quadratic curvature scalars such as the Kretschmann scalar have units  $L^{-4}$ . A corresponding inverse-square magnitude is

$$\kappa_K := \sqrt{|R_{abcd}R^{abcd}|}, \quad [\kappa_K] = L^{-2}. \quad (65)$$

The normalized scalar  $\ell_P^2 \kappa_K$  is therefore dimensionless on a four-dimensional spacetime without making that spacetime two-dimensional.

**Proposition 6.1** (Normalization non-entailment). *For any smooth  $n$ -dimensional manifold carrying a declared curvature magnitude  $\mathcal{K}$  of units  $L^{-2}$ ,*

$$\ell^2 \mathcal{K} \in \mathbb{R} \quad (66)$$

*is a dimensionless scalar. The exponent two determines the length weight of the normalization and does not determine  $n$ .*

*Proof.* The statement follows from dimensional homogeneity:  $[\ell^2 \mathcal{K}] = L^2 L^{-2} = 1$ . The topological dimension  $n$  does not occur in this unit equation. It enters through the domain of  $\mathcal{K}$ , the tensor contractions used to define it, and the integration measure if one forms an action or global functional.  $\square$

## 6.2 Local threshold versus integrated functional

A local normalized curvature scalar such as Eq. (64) should be distinguished from an integrated functional. On a three-dimensional Shadow domain one may define, for example,

$$\mathcal{V}_\Pi[\Omega] := \int_{\Omega \subset \Sigma^3} \mathcal{K}_\Pi^{3/2} dV_g, \quad (67)$$

which is dimensionless because  $[\mathcal{K}_\Pi^{3/2}] = L^{-3}$  and  $[dV_g] = L^3$ . Equation (67) makes the carrier dimension explicit through the measure. It complements rather than replaces the local normalization condition  $\mathcal{N}_P[\mathcal{K}_\Pi] = 1$ . Neither condition is identified with  $\mathcal{B}_\Pi = 1$  unless a source-to-Shadow bridge is derived.

## 6.3 The Planck scale is not itself a geometric object

The geometric types are therefore

$$\begin{aligned} \ell_P &: \text{scalar scale,} \\ \Omega_\Sigma &: \text{three-dimensional carrier domain,} \\ \mathcal{H}_\Sigma &: \text{nonempty two-dimensional regular boundary.} \end{aligned} \quad (68)$$

Only after Definition 4.1 is satisfied may  $\Omega_\Sigma$  and  $\mathcal{H}_\Sigma$  be denoted  $\Omega_\Pi$  and  $\mathcal{H}_\Pi$ . The scale, domain, and boundary are distinct mathematical objects.

## 7 Effective Ultraviolet Dimensional Reduction: The Technical Claim

### 7.1 Dimension is estimator-relative

The quantum-gravity literature does not use one universal notion of dimension. Carlip’s review is explicit that different estimators answer different physical questions and can disagree in nonclassical regimes [6]. Table 1 records the distinctions needed here.

Table 1: Dimension concepts relevant to the Planck regime.

| Quantity                               | Typical definition                                                                              | What it measures                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Topological dimension $d_{\text{top}}$ | Dimension of local coordinate charts or rank of the carrier                                     | Number of independent local directions in the underlying topological manifold.                     |
| Hausdorff dimension $d_{\text{H}}$     | $\mu(B_r) \sim r^{d_{\text{H}}}$                                                                | Scaling of measure or volume with radius.                                                          |
| Spectral dimension $d_{\text{s}}$      | $P(\sigma_{\text{D}}) \sim \sigma_{\text{D}}^{-d_{\text{s}}/2}$ , or its logarithmic derivative | Scaling seen by a specified diffusion process or positive operator.                                |
| Walk dimension $d_{\text{w}}$          | $\langle r^2 \rangle \sim \sigma_{\text{D}}^{2/d_{\text{w}}}$                                   | Anomalous spreading rate of the probe.                                                             |
| Threshold dimension                    | $\dim X^{-1}(\Gamma)$ under transversality                                                      | Topological dimension of a regular level-set preimage.                                             |
| Boundary/dual-theory dimension         | Dimension of a horizon section or auxiliary dual theory                                         | Geometry of a boundary or representation, not automatically the dimension of the represented bulk. |

Agreement among these quantities is a property of special geometries and operators, not a definition. In an ordinary smooth  $d$ -dimensional Riemannian manifold with the standard Laplace–Beltrami operator, the short-time spectral and topological dimensions agree. Quantum ensembles, anisotropic dispersion, nonlocal operators, discreteness, and fractal measures can break that equality.

### 7.2 Spectral dimension from a heat kernel

Let  $\mathcal{D}$  be a nonnegative self-adjoint diffusion generator on a declared state or function space. Its heat kernel satisfies

$$\partial_{\sigma_{\text{D}}} K(x, x'; \sigma_{\text{D}}) = -\mathcal{D}K(x, x'; \sigma_{\text{D}}), \quad K(x, x'; 0) = \delta(x, x'), \quad (69)$$

with boundary conditions and measure included in the definition of  $\mathcal{D}$ . For finite volume  $V$ , the average return probability is

$$P(\sigma_{\text{D}}) = \frac{1}{V} \int d\mu(x) K(x, x; \sigma_{\text{D}}), \quad (70)$$

and the scale-dependent spectral dimension is

$$d_{\text{s}}(\sigma_{\text{D}}) = -2 \frac{d \log P(\sigma_{\text{D}})}{d \log \sigma_{\text{D}}}. \quad (71)$$

For a smooth  $d$ -dimensional Riemannian manifold and an ordinary second-order Laplace-type operator,

$$P(\sigma_D) \sim (4\pi\sigma_D)^{-d/2} [1 + O(\sigma_D)] \quad (\sigma_D \downarrow 0), \quad (72)$$

so  $d_s \rightarrow d$  [21]. A different ultraviolet value therefore reports a change in the diffusion generator, dispersion relation, measure, ensemble average, or microscopic geometry sampled by the probe. It is a real analytic result, but its type is fixed by Eqs. (69)–(71).

### 7.3 The established ultraviolet results

In causal dynamical triangulations, Ambjørn, Jurkiewicz, and Loll fitted their measured spectral dimension over the simulated diffusion range by

$$D_S(\sigma_D) = 4.02 - \frac{119}{54 + \sigma_D}, \quad (73)$$

with reported asymptotic estimates

$$D_S(\infty) = 4.02 \pm 0.10, \quad D_S(0) = 1.80 \pm 0.25. \quad (74)$$

The result is a numerical, scale-dependent spectral dimension of the CDT ensemble, not a measurement of the topological dimension of each simplicial carrier [3].

Hořava extended the same probe to a smooth spacetime with anisotropic scaling exponent  $z$ . In  $D + 1$  dimensions he obtained

$$d_s = 1 + \frac{D}{z}. \quad (75)$$

For  $D = 3$ , the ultraviolet fixed point  $z = 3$  gives  $d_s = 2$ , while the infrared value  $z = 1$  gives  $d_s = 4$  [4]. This example directly separates ultraviolet spectral scaling from topological dimension because its topology remains smooth and four-dimensional. The reduction is carried by anisotropic propagation. Hořava explicitly notes that the observed spectral flow need not indicate a change in topological dimension.

Asymptotic-safety calculations provide a third route: the renormalization-group improved diffusion operator yields  $d_s = 2$  in the microscopic regime and  $d_s = 4$  macroscopically [5]. Causal-set analyses furnish still other effective-dimension estimators and illustrate that the behavior can depend on the selected probe and nonlocal operator [6, 7]. The convergence around two is therefore a genuine cross-programme pattern, but the common numerical value does not make the underlying mathematical mechanisms identical.

### 7.4 Two inequivalent appearances of the number two

**Proposition 7.1** (Spectral two and threshold two are distinct). *The statements*

$$d_{\text{top}}(\mathcal{H}_\Sigma) = 2 \quad (76)$$

and

$$\lim_{\sigma_D \downarrow 0} d_s(\sigma_D) = 2 \quad (77)$$

are logically independent unless a separate theorem identifies the diffusion problem in Eq. (77) with an operator intrinsically supported on the threshold  $\mathcal{H}_\Sigma$ .

*Proof.* For one direction, let  $\Omega$  be a smooth three-dimensional ball. Its boundary is a two-sphere, so  $d_{\text{top}}(\partial\Omega) = 2$ , while the short-time bulk heat trace of the ordinary Laplacian has leading exponent three. Thus Eq. (76) does not imply Eq. (77). For the converse, Hořava’s smooth topologically four-dimensional construction has  $d_s \rightarrow 2$  in the ultraviolet because of anisotropic scaling, without a topological rank loss. Thus Eq. (77) does not imply that the carrier or bulk has topological dimension two.  $\square$

Theorem 4.2 and Proposition 7.1 therefore locate two different mathematical objects:

$$\underbrace{\mathcal{H}_\Sigma^2}_{\text{codimension-one threshold}} \neq \underbrace{d_s(\sigma_D) \rightarrow 2}_{\text{ultraviolet diffusion scaling}}. \quad (78)$$

The first is a submanifold. The second is a function of resolution and of a declared operator.

## 7.5 When could the two notions be physically connected?

A bridge is possible, but it must be earned. To identify ultraviolet spectral reduction with localization on  $\mathcal{H}_\Sigma$ , a theory would need to show at least that

- (i) the ultraviolet diffusion generator is derived from the same admissibility/readout architecture that defines  $\mathcal{B}_\Sigma$ ;
- (ii) its physical domain becomes intrinsically supported on  $\mathcal{H}_\Sigma$ , or normal/bulk modes become gauge, infinitely costly, or operationally null;
- (iii) the heat trace used to define  $d_s$  is the trace of that physical operator rather than an arbitrarily selected auxiliary Laplacian;
- (iv) the resulting flow is stable under the declared class of physically equivalent selectors and boundary conditions;
- (v) a measurable propagation, correlation, or density-of-states consequence distinguishes this mechanism from alternative operators that also yield  $d_s \rightarrow 2$ .

Without these conditions, numerical equality at two is not object identity.

Within TCGS–SEQUENTION, spectral dimension should therefore be classified as a Shadow-level analytic readout. If  $d_s(\sigma_D)$  is preserved under the selector-equivalence groupoid for a fixed physical diffusion operator, it can be a relative invariant. If it changes with the probe, operator, carrier, or boundary sector, it remains a map-dependent but physically meaningful quantity. The framework can accommodate ultraviolet  $d_s \simeq 2$  while preserving a topologically three-dimensional Shadow carrier, but it does not predict the flow until the diffusion generator is derived from the source-to-readout map.

## 8 Black-Hole Area Laws and Holographic Descriptions

Black-hole thermodynamics and holographic dualities introduce lower-dimensional structures of a type distinct from spectral flow and carrier topology. The mature literature formulates entropy bounds, boundary descriptions, and dualities through explicit mathematical dictionaries [22–25]. Their inferential content is separated here from the ultraviolet spectral-dimension results of Section 7.

### 8.1 What the area law says

For a stationary black hole in semiclassical gravity, the Bekenstein–Hawking entropy is

$$S_{\text{BH}} = \frac{k_{\text{B}} A}{4\ell_{\text{P}}^2}. \quad (79)$$

The result establishes that black-hole entropy scales with the area of an appropriate horizon cross-section rather than with a naive bulk volume [26, 27]. Equation (79) does not identify that cross-section with the entire physical domain. A regular horizon section is codimension two in four-dimensional spacetime and codimension one in a spatial slice. Its area can control state counting while the bulk geometry still defines the horizon, the causal structure, and the field equations.

### 8.2 Encoding, duality, and ontology

Three statements must be separated:

$$\text{boundary data can constrain or encode bulk observables,} \quad (80)$$

$$\text{two theories can be dual descriptions of one physical content,} \quad (81)$$

$$\text{the represented physical carrier has lower topological dimension.} \quad (82)$$

The first two do not entail the third. Even an exact duality establishes an equivalence of theories or observables under a specified dictionary; whether one presentation is ontologically privileged is an additional interpretive question. In the TCGS–SEQUENTIONS vocabulary, 4D-Counterduction is not assumed to be holographic: the Shadow need not be a literal boundary, no equality of partition functions is postulated, and no globally invertible source reconstruction is assumed [8].

The categorical safeguard is

$$\text{area scaling / lower-dimensional encoding} \neq \text{bulk topological collapse.} \quad (83)$$

This conclusion is compatible with, rather than hostile to, technical holographic theories in their proper domains.

### 8.3 What would establish topological collapse

A physically meaningful claim that the carrier itself becomes two-dimensional would require a result of a different type from either an area law or a spectral flow. Examples include

- (i) a theorem that the local carrier rank drops,  $\text{rank}(dX) < 3$ , on a specified regime;
- (ii) a topology-changing construction in which the relevant state space ceases to admit three-dimensional local charts;

- (iii) a dynamical or constraint analysis showing that the normal degree is gauge, nondynamical, or absent from the physical phase space;
- (iv) an observable-algebra theorem proving that every physically distinguishable local degree of freedom is represented on a two-dimensional carrier and that putative bulk directions carry no independent causal or operational content.

A running spectral dimension is not a “two-dimensional tangent distribution” and should not be described as one. It is the scaling of a heat trace or return probability for a specified operator. Neither  $A/\ell_P^2$ ,  $\ell_P^2 \mathcal{K}_\Pi$ , nor  $d_s \rightarrow 2$  supplies conditions (i)–(iv) by itself.

## 9 The Signature and Causal-Structure Problem

### 9.1 The source signature is not fixed by the boundary theorem

Theorem 4.2 is differential-topological. It uses smoothness, rank, regularity, and transversality. It does not require a Riemannian or Lorentzian source metric. This is an advantage for the dimensional result and a limitation for the physical ontology.

Let  $G$  be a nondegenerate source metric if a metric realization is chosen. No source signature is fixed here:

$$\text{sig}(G) = (+, +, +, +) \quad \text{or} \quad \text{sig}(G) = (-, +, +, +). \quad (84)$$

The framework’s source-map discipline places smoothness class, topology, metric signature, field representation, and operator realization inside the provisional scientific representation until they are separately derived [10]. Signature is therefore an explicit closure condition rather than an inherited property of the source notation.

The spectral-dimension calculation does not fix this choice. Its diffusion generator is positive and is commonly defined on a Euclideanized geometry or an explicitly elliptic operator; Hořava’s derivation, for example, first constructs the appropriate anisotropic diffusion problem [4, 21]. A heat-kernel exponent therefore cannot be substituted for a Lorentzian-signature theorem or for the derivation of physical null cones.

### 9.2 The two branches of the problem

The paper therefore distinguishes

$$G \text{ (source geometry, signature open)} \quad \text{from} \quad g_{\mu\nu}^{\text{eff}} \text{ (effective Lorentzian Shadow metric)}. \quad (85)$$

The invariant speed  $c$  belongs to the null structure of the effective Shadow description. It is not, without a separate derivation, a velocity of 4D-Counterduction or a primitive operation in Counterspace.

### 9.3 What BSW/ADM does and does not supply

The Baierlein–Sharp–Wheeler action and ADM constraint formalism show that general relativity can be represented relationally through constrained families of three-geometries [28–30]. Under the usual locality, supermetric, potential, and closure assumptions, a Lorentzian four-metric can

Table 2: Consequences of the unresolved source-signature choice.

| Source-map choice               | Immediate advantage                                                         | Unresolved burden                                                                                                                                                            |
|---------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lorentzian $G$                  | Null cones and causal classification are available at source-map level.     | A mathematical discriminator is required to show why $\mathcal{C}^4$ is not simply ordinary spacetime with a relabelled foliation; “counter-spatial” cannot remain semantic. |
| Riemannian $G$                  | The fourth source degree is manifestly non-temporal in metric signature.    | The effective Lorentzian metric, null structure, finite invariant speed $c$ , and matter propagation in the Shadow must be derived from the readout architecture.            |
| Signature-free source structure | The preimage and codimension results remain valid with minimal assumptions. | No causal or propagation result follows until a metric or equivalent causal structure is constructed.                                                                        |

be reconstructed at the level of the relativistic comparison theory. This is important because it demonstrates that a freely relabelable ordering parameter need not be an external Newtonian time.

It does not, by itself, prove that the reconstructed Lorentzian spacetime is a readout of a distinct non-temporal source manifold. Nor does it determine  $\text{sig}(G)$  in Eq. (85). The source-to-causality bridge remains:

$$(\mathcal{C}^4, G, \Psi; \text{Ctd}) \xrightarrow{\text{open}} (M^4, g_{\mu\nu}^{\text{eff}}, \text{null cones}, c). \quad (86)$$

A future signature paper must construct this arrow and show why its output is Lorentzian with the observed invariant speed and causal structure.

#### 9.4 Relational time is not unique evidence for Counterspace

Page and Wootters showed how dynamics can be described through correlations among stationary observables without an external time parameter [31]. Barontini’s 2026 cold-atom experiment quantitatively tested an internal relational-time construction in an isolated Bose–Einstein-condensate system [32]. These results are relevant to the anti-reification of time: operational ordering can be generated relationally within a theory whose global description does not contain an external clock.

They also impose a strict inference limit. Relational time can be formulated inside a conventional quantum or Lorentzian setting. Therefore

$$\text{non-fundamental external time} \not\Rightarrow \text{four-dimensional Counterspace}. \quad (87)$$

The distinctive TCGS–SEQUENTIONS claim is not merely that time can be relational. It is the additional source–Shadow architecture, the exclusion of the ordering label from source dimensionality, and the conditional minimal-normal-degree theorem. Those additional premises must stand on their own.

## 10 A Worked Schwarzschild Shadow-Spacetime Calibration

A concrete curvature calibration illustrates the distinction between a scale-normalized Shadow diagnostic and a source-preimage threshold. The calculation is performed on the standard effective Lorentzian Schwarzschild spacetime  $(\mathcal{M}_{\text{Schw}}^4, g_{\text{Schw}})$  with spatial embedding

$$i_t : \Sigma_t^3 \hookrightarrow \mathcal{M}_{\text{Schw}}^4. \quad (88)$$

This is a Shadow-spacetime construction. It is not the source immersion  $X : \Sigma^3 \hookrightarrow \mathcal{C}^4$ , and the scalar used below is not  $\mathcal{B}_\Pi = X^* \mathcal{B}_C$ .

### 10.1 Source pullback versus a functional of Shadow geometry

Two operations that look similar in compressed notation must be separated:

$$\underbrace{X^*(F_C[G, \Psi])}_{\text{pullback of a source scalar}} \quad \text{and} \quad \underbrace{F_\Sigma[X^*G, X^*\Psi]}_{\text{functional formed from induced data}}. \quad (89)$$

They are not equal in general. For a codimension-one immersion with unit normal  $n$ ,  $G(n, n) = \varepsilon \in \{+1, -1\}$ , and second fundamental form  $K_{ij}$ , the Gauss equation gives

$$R_{ijkl}^\Sigma = (X^* R^C)_{ijkl} + \varepsilon (K_{ik} K_{jl} - K_{il} K_{jk}), \quad (90)$$

so curvature functionals of the induced metric generally contain extrinsic terms and do not commute with pullback [33]. Normal curvature components supply additional information not contained in the intrinsic carrier tensor.

The Schwarzschild diagnostic is a third, explicitly Shadow-level object:

$$\hat{\mathcal{B}}_K := \ell_P^2 \sqrt{R_{\mu\nu\rho\sigma}[g_{\text{Schw}}] R^{\mu\nu\rho\sigma}[g_{\text{Schw}}]} : \mathcal{M}_{\text{Schw}}^4 \setminus \{r = 0\} \longrightarrow \mathbb{R}. \quad (91)$$

Let

$$\hat{\Gamma}_K := \hat{\mathcal{B}}_K^{-1}(1), \quad \mathcal{S}_K(t) := i_t^{-1}(\hat{\Gamma}_K). \quad (92)$$

The set  $\mathcal{S}_K(t)$  is a level-set sphere in a spatial slice of the effective Schwarzschild spacetime. To identify it with the source-preimage threshold  $\mathcal{H}_\Sigma$ , a future source realization must construct a comparison diffeomorphism  $J : \Sigma^3 \rightarrow \Sigma_t^3$  and prove

$$X^* \mathcal{B}_C = J^* i_t^* \hat{\mathcal{B}}_K \quad (93)$$

together with the relevant transversality and selector-naturality conditions. No such bridge is assumed here. Accordingly,

$$\mathcal{S}_K(t) = \mathcal{H}_\Sigma \quad \text{requires Eq. (93)}. \quad (94)$$

Absent that bridge, the two level sets remain distinctly typed objects whose identification is unestablished.

## 10.2 Why the Kretschmann choice is forced in the vacuum exterior

For the Schwarzschild exterior,

$$R = 0, \quad R_{ab} = 0, \quad R_{ab}R^{ab} = 0. \quad (95)$$

Thus the Ricci scalar and Ricci-tensor square cannot define a nontrivial local curvature threshold. At quadratic order in the Riemann tensor, the Kretschmann scalar is the available nonzero parity-even scalar invariant in this geometry. Its use here is therefore not an arbitrary selection among several nonzero quadratic candidates.

For Schwarzschild mass  $M$  [34, 35],

$$K_{\text{Schw}} := R_{abcd}R^{abcd} = \frac{48G^2M^2}{c^4r^6}. \quad (96)$$

Define the inverse-square curvature magnitude

$$\kappa_K(r; M) := \sqrt{K_{\text{Schw}}} = \frac{4\sqrt{3}GM}{c^2r^3}, \quad (97)$$

so the Shadow-spacetime diagnostic is

$$\hat{\mathcal{B}}_K(r; M) = \ell_P^2 \kappa_K(r; M). \quad (98)$$

Because  $\hat{\mathcal{B}}_K$  is a scalar constructed from  $g_{\text{Schw}}$ , every Schwarzschild isometry preserves it. In particular, for the stationary Killing field  $\partial_t$  and every rotational Killing field  $\zeta \in \mathfrak{so}(3)$ ,

$$\mathcal{L}_{\partial_t} \hat{\mathcal{B}}_K = 0, \quad \mathcal{L}_{\zeta} \hat{\mathcal{B}}_K = 0. \quad (99)$$

Equation (99) is a worked Shadow-spacetime instance of a first-integral condition of the form in Eq. (54). It supports the mathematical plausibility of orbit-invariant threshold fields while supplying no identification of  $\hat{\mathcal{B}}_K$  with the source field  $\mathcal{B}_C$ .

## 10.3 The general threshold sphere

Using  $r_s = 2GM/c^2$ , Eq. (98) becomes

$$\hat{\mathcal{B}}_K(r; M) = \frac{2\sqrt{3}\ell_P^2 r_s}{r^3}. \quad (100)$$

The level  $\hat{\mathcal{B}}_K = 1$  therefore occurs at

$$r_K^3 = 2\sqrt{3}\ell_P^2 r_s, \quad (101)$$

$$\boxed{r_K = (2\sqrt{3})^{1/3} (\ell_P^2 r_s)^{1/3} \simeq 1.5131 (\ell_P^2 r_s)^{1/3}} \quad (102)$$

or, equivalently,

$$r_K = \ell_P \left( 4\sqrt{3} \frac{M}{m_P} \right)^{1/3}. \quad (103)$$

For every positive Schwarzschild mass, the declared Shadow-spacetime curvature diagnostic therefore defines

$$\mathcal{S}_K(t) \cong S_{r_K}^2 \quad (104)$$

on a regular spatial slice. The sphere is nested in a three-dimensional radial regime. This is a closed level-set statement inside the Schwarzschild map; it is neither a claim that the full Planck region is two-dimensional nor a realization of  $\mathcal{H}_\Sigma = X^{-1}(\Gamma)$ .

The ratio to the horizon is

$$\frac{r_K}{r_s} = \left[ \frac{\sqrt{3}}{2} \left( \frac{m_P}{M} \right)^2 \right]^{1/3}. \quad (105)$$

Thus  $r_K < r_s$  for  $M > M_\star$ ,  $r_K = r_s$  at  $M = M_\star$ , and  $r_K > r_s$  for  $M < M_\star$ . For a solar-mass Schwarzschild radius  $r_{s,\odot} \simeq 2.95$  km, Eq. (102) gives

$$r_K(M_\odot) \simeq 1.39 \times 10^{-22} \text{ m}, \quad \frac{r_K}{r_{s,\odot}} \simeq 4.70 \times 10^{-26}. \quad (106)$$

This illustrates the standard statement that curvature becomes Planckian deep toward the Schwarzschild singularity for a macroscopic black hole, not at its horizon.

#### 10.4 The special horizon-coincidence mass

At the horizon,

$$\hat{\mathcal{B}}_K(r_s; M) = 2\sqrt{3} \left( \frac{\ell_P}{r_s} \right)^2 = \frac{\sqrt{3}}{2} \left( \frac{m_P}{M} \right)^2. \quad (107)$$

At  $M = m_P$ , this is  $\sqrt{3}/2 \simeq 0.8660$ . The threshold sphere coincides with the horizon when

$$M_\star = \sqrt{\frac{\sqrt{3}}{2}} m_P = \frac{3^{1/4}}{\sqrt{2}} m_P \simeq 0.9306 m_P, \quad (108)$$

$$r_{s,\star} = 2\ell_P \frac{M_\star}{m_P} \simeq 1.8612 \ell_P. \quad (109)$$

Equations (108)–(109) are therefore a special case of the general threshold equation, obtained by imposing the additional condition  $r_K = r_s$ .

#### 10.5 Comparison with the Compton–Schwarzschild intersection

Using the reduced Compton wavelength

$$\bar{\lambda}_C = \frac{\hbar}{Mc} = \ell_P \frac{m_P}{M}, \quad (110)$$

and equating it to  $r_s$  gives

$$M = \frac{m_P}{\sqrt{2}}, \quad r = \sqrt{2} \ell_P. \quad (111)$$

Alternative conventions using the gravitational radius  $GM/c^2$  or the unreduced Compton wavelength change the order-unity coefficients. The horizon-coincidence point in Eqs. (108)–(109) lies in the same order-Planck regime but is not identical to Eq. (111). Dimensional analysis fixes the scale up to dimensionless coefficients; a declared curvature diagnostic fixes a particular coefficient.

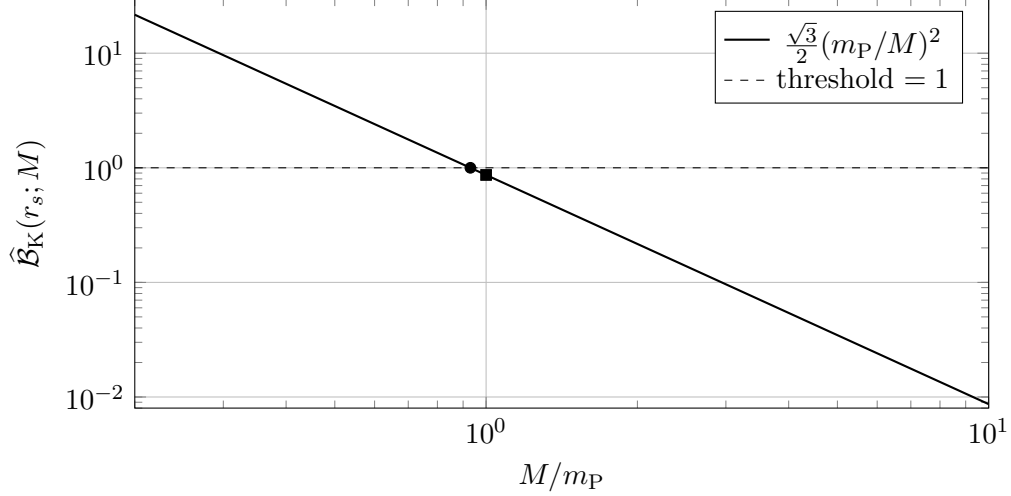


Figure 3: Horizon value of the declared Schwarzschild Shadow-spacetime curvature diagnostic. The crossing at  $M/m_P \simeq 0.9306$  is the special point at which the level-set sphere  $r = r_K$  coincides with  $r = r_s$ . The plot does not identify this sphere with the source-preimage threshold  $\mathcal{H}_\Sigma$ .

## 10.6 Where invariant-choice dependence can actually be demonstrated

Schwarzschild alone does not demonstrate dependence on the choice among nonzero quadratic invariants, because Eq. (95) removes the Ricci candidates. To exhibit the issue, consider a four-dimensional maximally symmetric non-Ricci-flat Shadow geometry satisfying  $R_{ab} = \Lambda g_{ab}$ . Then

$$R = 4\Lambda, \quad R_{ab}R^{ab} = 4\Lambda^2, \quad R_{abcd}R^{abcd} = \frac{8}{3}\Lambda^2. \quad (112)$$

Three individually natural dimensionless thresholds give

$$\ell_P^2 |R| = 1 \quad \Rightarrow \quad |\Lambda| \ell_P^2 = \frac{1}{4}, \quad (113)$$

$$\ell_P^2 \sqrt{R_{ab}R^{ab}} = 1 \quad \Rightarrow \quad |\Lambda| \ell_P^2 = \frac{1}{2}, \quad (114)$$

$$\ell_P^2 \sqrt{R_{abcd}R^{abcd}} = 1 \quad \Rightarrow \quad |\Lambda| \ell_P^2 = \sqrt{\frac{3}{8}}. \quad (115)$$

The thresholds agree at the order-Planck level but differ by order-unity coefficients. This example, not the Ricci-flat Schwarzschild exterior, demonstrates diagnostic dependence. A completed source theory must derive which Shadow invariant, if any, is the readout of  $\mathcal{B}_C$  before the calculation can be promoted to a source threshold.

The worked calculations support five controlled conclusions. A declared Shadow curvature invariant produces an executable level set. The Schwarzschild level is a two-sphere embedded in a three-dimensional spatial regime for every mass. Its coincidence with the horizon occurs only near the Planck mass. Order-unity diagnostic dependence is real when several nonzero curvature invariants are available. Finally, none of these statements identifies  $\mathcal{S}_K(t)$  with  $\mathcal{H}_\Sigma$ ; that identification remains the open bridge in Eq. (93).

## 11 Planck Time, Operational Clocks, and Gauge Labels

The standard statement

$$t_P = \frac{\ell_P}{c} \quad (116)$$

remains operationally meaningful: it is the light-crossing interval associated with one Planck length in a Lorentzian description. It does not follow that time is composed of indivisible intervals  $t_P$ .

Within the TCGS-SEQUENTION type discipline, three objects are distinguished:

$$\mathcal{F} = \{X_\lambda : \Sigma^3 \hookrightarrow \mathcal{C}^4\}, \quad \lambda \sim h(\lambda), \quad \tau_{\text{op}} = \mathcal{T}[\text{clocks, correlations, records}]. \quad (117)$$

The foliation or admissible family has geometric content; its bare label is gauge under admissible relabelling; and operational time is a physical Shadow functional constructed from clocks and correlations [8, 36]. Thus

$$\mathcal{F} \neq \lambda \neq \tau_{\text{op}}, \quad \tau_{\text{op}} \notin \text{Dim}(\mathcal{C}^4). \quad (118)$$

Page–Wootters and Barontini establish that internal correlations can order dynamics without an external fundamental clock [31, 32]. They support the narrow proposition that clock order and ontic temporal substance are not identical. The further TCGS-SEQUENTION proposition that the source possesses a non-temporal fourth degree remains conditional on the source-completion premises and on the unresolved causal bridge in Eq. (86).

## 12 Projection-Admissibility Rescaling Ansatz

Consider the dimensionless substitution

$$G \longmapsto G\mathcal{A}_\Pi, \quad \mathcal{A}_\Pi > 0. \quad (119)$$

Applied to the Planck generator, it defines

$$Q_\Pi(m, \ell, \tau) = \hbar^{\frac{m+\ell+\tau}{2}} c^{\frac{m-3\ell-5\tau}{2}} (G\mathcal{A}_\Pi)^{\frac{\ell+\tau-m}{2}}, \quad (120)$$

and therefore the exact identity

$$Q_\Pi(m, \ell, \tau) = Q_P(m, \ell, \tau) \mathcal{A}_\Pi^{\frac{\ell+\tau-m}{2}}. \quad (121)$$

Equation (121) is a one-parameter rescaling controlled by the exponent of  $G$  in the corresponding Planck monomial.

For length and energy,

$$\ell_\Pi = \ell_P \mathcal{A}_\Pi^{1/2}, \quad E_\Pi = E_P \mathcal{A}_\Pi^{-1/2}, \quad (122)$$

so

$$\ell_\Pi E_\Pi = \ell_P E_P = \hbar c. \quad (123)$$

This equality is an algebraic cancellation already present in the ordinary Planck units. It does not provide an independent invariant of projection geometry.

**Proposition 12.1** (Physical status of the  $\mathcal{A}_\Pi$  ansatz). *The substitution in Eq. (119) is dimensionally consistent but becomes a physical deformation only if all of the following are supplied:*

- (i) an independently specified geometric functional  $\mathcal{A}_\Pi = F[I_1, I_2, \dots]$ ;
- (ii) a declared source or Shadow domain on which the invariants are defined;
- (iii) a non-circular parameter rule fixed before comparison with target data;
- (iv) at least one observable consequence not reducible to Eq. (121);
- (v) compatibility with low-energy gravitational, Lorentz, and equivalence-principle constraints.

The complete algebraic consequences are recorded in Appendix B. No theorem or calibration in the main text depends on  $\mathcal{A}_\Pi$ .

### 13 Relation to Quantum-Gravity Programmes

Two questions have different mathematical content. The first asks whether a quantum-gravity model derives an effective ultraviolet dimension near two. Several models do, as summarized in Section 7. The second asks whether the physical carrier thereby becomes topologically two-dimensional. That stronger conclusion does not follow automatically and is explicitly denied in Hořava’s smooth anisotropic example [4].

Loop quantum gravity derives discrete spectra for area and volume operators in its own formal construction [37, 38]. The coexistence of area and volume spectra already blocks the naive inference from area quantization to a two-dimensional carrier, while the boost analysis of Rovelli and Speziale shows why a minimum geometric eigenvalue need not behave as a Lorentz-contracted classical tile [16]. String theory introduces extended-object scales, thermodynamic dimensional behavior, and dualities. Causal sets introduce locally finite order, Lorentz-compatible random sprinklings, and nonlocal diffusion operators rather than a regular spatial lattice [7, 17, 18, 39]. Deformed-special-relativity models illustrate a separate route in which the boost law itself is modified to accommodate an observer-independent Planckian scale [19]. Effective field theory treats general relativity as a controlled low-energy expansion without presuming that  $\ell_P$  is a literal lattice spacing [11]. Asymptotic safety and CDT produce running spectral dimensions through different calculational structures [3, 5]. Holography provides powerful bulk–boundary dictionaries, but its dimensional relation is neither the same as spectral flow nor the same as topological rank collapse.

The framework-independent conclusions are therefore two paired non-entailments:

$$\left. \begin{array}{l} \ell_P^2 \times \text{curvature is dimensionless,} \\ S_{\text{BH}} \propto A/\ell_P^2, \\ \text{bulk observables admit a boundary description} \end{array} \right\} \not\Rightarrow d_s \rightarrow 2, \quad (124)$$

$$d_s \rightarrow 2 \not\Rightarrow d_{\text{top}}(\text{bulk}) = 2. \quad (125)$$

Equation (124) prevents Planck or holographic notation from manufacturing a diffusion result. Equation (125) prevents a valid diffusion result from being promoted beyond its mathematical type.

The positive TCGS–SEQUENTION interpretation is conditional. The Planck threshold is read as a boundary of autonomous three-dimensional Shadow description relative to a four-dimensional source architecture, while a possible ultraviolet spectral flow would describe how a source-conditioned Shadow propagation operator is read at short resolution. The preimage theorem makes the first claim coherent. The framework does not derive the second until it constructs  $\mathcal{D}$  from the

ECL/Counterduction chain and proves selector stability. Neither result by itself proves that nature realizes the source ontology.

## 14 Scope of Results and Decisive Closure Conditions

Tables 3 and 4 separate established results, non-entailments, conditional constructions, and open bridges. The classification is part of the formal architecture: results from dimensional analysis, differential topology, diffusion theory, and Shadow-spacetime curvature are not assigned source-ontological force beyond their stated domains.

**Criterion 14.1** (Source field, domain, and crossing). Specify a dimensionless  $\mathcal{B}_C$  from a concrete source geometry and readability map, prove regularity of  $\Gamma$ , construct the physical immersion  $X$ , establish  $\mathcal{H}_\Sigma \neq \emptyset$ , and verify Eq. (38). A carrier contained in  $\Gamma$  is threshold-saturated rather than bounded by a two-dimensional preimage.

**Criterion 14.2** (Planck-normalization bridge). Either derive an intrinsic source normalization package satisfying Eq. (25), or declare a readout-calibrated normalization satisfying Eqs. (27)–(28). In the calibrated route, record the reference selector, calibration domain, uncertainty, and the comparison action required by Criterion 14.3. Calibration and transport must then be solved jointly through Proposition 5.6; they are not independent adjustable bridges. No source-level Planck claim may use  $c$  or  $G$  before the causal and gravitational readout maps that define them have been supplied.

**Criterion 14.3** (Selector groupoid and leaf-changing lift). Construct  $\text{Sel}_\Xi$  without using the threshold as the definition of equivalence. For every leaf-changing arrow, specify  $\varphi_\kappa$  and a source lift  $\mathbf{f}_\kappa$  satisfying Eq. (51). For a one-parameter family, derive its generator  $\xi$ , test  $\mathcal{L}_\xi \mathcal{B}_C = 0$ , and, when the readout-calibrated route of Criterion 14.2 is used, verify transversality to  $\mathcal{L}_0$  and the orbit-sweeping hypothesis of Eq. (56). Under those hypotheses, Proposition 5.6 uniquely propagates the calibrated field; failure of the physical comparison flow blocks the construction before threshold transport is asserted.

Criteria 14.2 and 14.3 therefore define one coupled initial-value problem. The physical architecture must supply the reference-leaf data and the comparison flow. Once those inputs satisfy the stated flow-box hypotheses, the source admissibility field is fixed uniquely on the swept region by Eq. (58).

**Criterion 14.4** (Source–Schwarzschild bridge). Derive the effective Lorentzian Shadow metric from the source architecture, specify  $J : \Sigma^3 \rightarrow \Sigma_t^3$ , and prove Eq. (93). Without that commuting relation,  $\mathcal{S}_K(t)$  remains a Shadow-spacetime calibration.

**Criterion 14.5** (Spectral bridge). Derive the physical diffusion generator  $\mathcal{D}$  from the source-to-readout architecture, define its domain and measure, compute  $d_s(\sigma_D)$ , and test stability along arrows of  $\text{Sel}_\Xi$ . A numerical limit near two is not assigned to  $\mathcal{H}_\Sigma$  without the support and normal-mode bridge.

**Criterion 14.6** (Signature-to-causality bridge). Fix the mathematical type of  $G$  or provide an alternative source causal structure. Derive the effective Lorentzian metric, null cones, invariant speed  $c$ , matter propagation, and observer comparison maps in the Shadow.

Table 3: Framework-independent result and scope matrix.

| Result or construction                                                            | Status                                     | Mathematical or physical scope                                                                                                     |
|-----------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Planck monomials for declared target dimensions                                   | Established                                | Equations (5)–(7); coefficients remain convention dependent.                                                                       |
| Ontological meaning of $\ell_P$ and $t_P$                                         | Underdetermined by dimensional analysis    | The monomials contain no lattice, topology, or discreteness theorem.                                                               |
| Regular rigid Planck lattice under undeformed Lorentz boosts                      | Excluded without additional structure      | Proposition 5.1; a preferred frame, broken symmetry, or deformed boost law would be required.                                      |
| Other forms of Planck-scale discreteness                                          | Model dependent                            | Operator spectra, causal sprinklings, finite records, representation cutoffs, and spectral flow have different mathematical types. |
| Ultraviolet $d_s \simeq 2$ in CDT, Hořava–Lifshitz gravity, and asymptotic safety | Established within the cited models        | Equations (73)–(75), with their declared diffusion operators and ensembles.                                                        |
| Relation between $d_s \rightarrow 2$ and carrier topology                         | Independent absent a bridge                | Proposition 7.1; smooth topologically four-dimensional examples exhibit spectral two.                                              |
| Relation between ultraviolet spectral two and a two-dimensional interface         | Independent absent a bridge                | Requires a common operator, support relation, selector transport, and observable discriminator.                                    |
| Regular source-interface preimage                                                 | Conditional standard theorem               | Theorem 4.2; transversality permits an empty preimage, and nonempty boundary components have codimension one.                      |
| Admissible carrier domain                                                         | Established under the theorem hypotheses   | $\Omega_\Sigma$ is a smooth three-manifold with boundary $\mathcal{H}_\Sigma$ , including the empty-boundary case.                 |
| Physical transversality and nonempty crossing                                     | Constructive requirement                   | Equation (38); genericity of arbitrary immersions does not identify the selected physical carrier.                                 |
| Scale content of Theorem 4.2                                                      | Scale free                                 | Planck interpretation requires Definition 4.1; the theorem itself contains no $\hbar$ , $c$ , or $G$ .                             |
| Ontological discrimination by the preimage theorem                                | None                                       | The result applies to any four-manifold satisfying its smoothness and transversality hypotheses.                                   |
| Curvature normalization $\ell_P^2 \mathcal{K}$                                    | Dimensionally valid, topology neutral      | The exponent compensates inverse-square curvature units and does not count manifold coordinates.                                   |
| Black-hole area law and holographic description                                   | Boundary or duality statements             | Bulk topological reduction requires an independent rank, topology, or physical-phase-space result.                                 |
| Schwarzschild unit-curvature sphere $\mathcal{S}_K(t)$                            | Established Shadow-spacetime calculation   | Equations (101)–(103).                                                                                                             |
| Schwarzschild isometry invariance of $\hat{\mathcal{B}}_K$                        | Established Shadow-level consistency check | Equation (99); stationary and rotational Killing flows preserve the diagnostic without identifying it with $\mathcal{B}_C$ .       |
| Identification $\mathcal{S}_K(t) \cong \mathcal{H}_\Sigma$                        | Open cross-level bridge                    | Requires Eq. (93), effective metric derivation, and compatible transversality.                                                     |
| Curvature-diagnostic dependence                                                   | Demonstrated outside Ricci-flat vacuum     | The maximally symmetric comparison supplies multiple nonzero invariants and distinct order-unity thresholds.                       |

Table 4: Conditional TCGS–SEQUENTION result and scope matrix.

| Framework element                                        | Status                                           | Basis or closure requirement                                                                                                                                                                       |
|----------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source–Shadow map types                                  | Fixed by definition                              | $X$ is an immersion, $X^*$ is pullback, and Ctd is the constitutive readout relation.                                                                                                              |
| Lower bound on source dimension                          | Conditional                                      | $\mathcal{P}_{\geq 4}$ yields $\dim C \geq 4$ under the declared completion premises.                                                                                                              |
| Exact source dimension four                              | Conditional parsimony result                     | $\mathcal{P}_{\min}$ excludes further independent source directions; it is not a consequence of rank arithmetic.                                                                                   |
| Non-temporal type of the additional direction            | Conditional typing result                        | $\mathcal{P}_{-t}$ excludes operational time and foliation labels from source dimensionality.                                                                                                      |
| Intrinsic-source Planck normalization                    | Open                                             | Requires Route (i) of Definition 4.1, including a source package whose readout yields $(\hbar, c, G)$ .                                                                                            |
| Readout-calibrated orbit extension                       | Conditional constructive result                  | Route (ii) supplies data on $\mathcal{L}_0$ ; Proposition 5.6 gives the unique extension on a flow-swept region when the comparison generator is transverse and its flow is a diffeomorphic sweep. |
| Source signature and Lorentzian Shadow causality         | Open and load bearing                            | Requires Eq. (86), including null cones, invariant speed, matter propagation, and observer maps.                                                                                                   |
| Planck-regime object                                     | Conditionally scalar and geometric               | The proposed object is a normalized admissibility level set, not a regular tessellation of rigid coordinate cells.                                                                                 |
| Selector equivalence                                     | Independently specified groupoid                 | $\text{Sel}_{\Xi}$ is fixed from metric, clock, matter-readout, and corridor data before threshold naturality is tested.                                                                           |
| Naturality under pure carrier relabeling                 | Automatic identity                               | Lemma 5.3; it supplies consistency but no field restriction.                                                                                                                                       |
| Physical leaf-changing comparison action                 | Open physical input with a closed transport step | Proposition 5.5 gives the orbit-invariance equation; once a transverse orbit-sweeping flow is constructed, Proposition 5.6 solves the calibrated extension by characteristics.                     |
| Source–Schwarzschild identification                      | Open                                             | Functional formation and pullback do not commute automatically; Eq. (93) is required.                                                                                                              |
| Relational operational time                              | Compatible but non-unique evidence               | Page–Wootters and Barontini realize relational ordering in architectures that do not entail Counterspace.                                                                                          |
| Projection-admissibility factor $\mathcal{A}_{\Pi}$      | Dimensionally consistent ansatz                  | Equation (121); physical content requires an independently derived functional and discriminator.                                                                                                   |
| Planck scales as limits of autonomous Shadow description | Conditional framework interpretation             | Requires normalization, physical immersion, crossing, signature, causal readout, selector transport, and source-to-observable closure.                                                             |

**Criterion 14.7** (Nontrivial Planck deformation). If  $\mathcal{A}_\Pi$  is retained, derive it from independently specified invariants and carry it through a concrete geometry to a bound or prediction not reducible to Eq. (121).

**Criterion 14.8** (Topological-degeneracy test). A lower-dimensional Planck carrier requires a declared rank-drop or topology-change set, a dynamical mechanism, and an operational consequence distinguishing it from spectral reduction and from an ordinary codimension-one boundary.

## 15 Discussion

Ultraviolet dimensional reduction is a technically established phenomenon in several quantum-gravity programmes, but the word “dimension” refers to different estimators. Spectral dimension is defined by a diffusion operator and its heat trace; topological dimension is a property of local charts or manifold rank; Hausdorff dimension concerns measure scaling; boundary dimension follows from codimension; and holographic dimension can refer to an auxiliary dual description. The convergence of several ultraviolet calculations near two is physically significant without forcing these categories to coincide.

The Lorentz-contraction calculation supplies a similarly typed conclusion. An undeformed Lorentz group is incompatible with a regular microscopic tessellation whose classical cells have coordinate edge  $\ell_P$  in every inertial frame. The conclusion concerns rigid regular cells, not every object described as discrete. Geometric-operator spectra, stochastic causal sets, deformed boosts, and spectral flow employ different mathematical carriers and transformation laws [16–19]. The covariance question must be posed for the actual observable or combinatorial structure.

The source-interface theorem is deliberately scale-free. Under regularity and transversality,  $\Omega_\Sigma$  is a three-manifold with boundary  $\mathcal{H}_\Sigma$ , and every nonempty boundary component is two-dimensional. The theorem also accommodates an empty threshold, because transversality is vacuous when the carrier misses  $\Gamma$ . Nonemptiness and crossing are therefore independent physical requirements. If the carrier is entirely contained in the interface, the threshold is three-dimensional; if contact is tangential, the preimage may be singular or stratified. These alternatives are not pathologies of the theorem but distinct geometric regimes.

Calling  $\mathcal{H}_\Sigma$  a Planck threshold introduces a further bridge. The theorem contains no  $\hbar$ ,  $c$ , or  $G$ . A source-level normalization must either derive quantities whose readout yields the observed Planck package or import the observed constants from a calibrated Shadow selector. The imported route can organize a cross-level diagnostic, but it cannot establish that the source intrinsically contains the constants used for its normalization. This limitation is especially important in TCGS–SEQUENTION, where  $c$  is assigned to the null structure of the effective Lorentzian Shadow metric. Planck normalization and causal reconstruction are consequently coupled closure problems.

Selector covariance divides into a formal identity and a physical condition. For a pure relabeling of one carrier, scalar naturality follows by associativity of composition and imposes no restriction on  $\mathcal{B}_C$ . For a leaf-changing comparison, Proposition 5.5 identifies the obstruction exactly: the field must be invariant along the lifted source comparison. A one-parameter lifted observer family therefore requires  $\mathcal{L}_\xi \mathcal{B}_C = 0$ .

The main constructive result of the source-field sector is Proposition 5.6. Under the readout-calibrated route, the measured Shadow diagnostic fixes  $b_0$  on one embedded reference leaf. If a physically derived comparison generator is transverse to that leaf and its flow sweeps the relevant

source region once, the transport equation  $\mathcal{L}_\xi \mathcal{B}_C = 0$  determines a unique extension by characteristics. Calibration and covariance are therefore one coupled initial-value problem rather than two unrelated open conditions. The remaining uncertainty lies in the physical derivation and global coverage of the comparison flow, not in the local extension method once those inputs are supplied.

The distinction between spectral and threshold dimension is then exact. Proposition 7.1 exhibits both non-implications. A three-ball has a two-dimensional boundary while its ordinary bulk spectral dimension is three. Hořava’s smooth topologically four-dimensional construction has ultraviolet  $d_s = 2$  without topological rank loss. Numerical equality at two does not identify the underlying objects. A bridge would require a diffusion operator derived from the same admissibility architecture, physical support on the threshold or elimination of normal modes, selector stability, and an observable discriminator.

The Schwarzschild calculation has a separate role. The scalar  $\hat{\mathcal{B}}_K$  is fully defined on an effective Lorentzian spacetime and produces a two-sphere for every positive mass. For a macroscopic black hole the sphere lies deep inside the horizon; the special mass  $M_\star$  marks the case in which the unit-curvature sphere and the horizon coincide. Equation (99) supplies an internal covariance check: the diagnostic is a first integral of the stationary and rotational Killing flows. This is a Shadow-level realization of orbit invariance, not a source-level identification. Because  $\hat{\mathcal{B}}_K$  is constructed from  $g_{\text{Schw}}$  while  $\mathcal{B}_\Sigma$  is the pullback of a source scalar, the two constructions are not interchangeable. The Gauss equation displays the extrinsic terms that obstruct commutation between pullback and curvature-functional formation. Equation (93) is therefore a substantive physical bridge, not a notational identification.

The Schwarzschild exterior also clarifies diagnostic choice. Ricci-flatness removes the Ricci scalar and Ricci-tensor square, making the Kretschmann scalar the available nonzero parity-even quadratic curvature invariant. A maximally symmetric non-vacuum geometry then shows how different nonzero curvature diagnostics produce different order-unity threshold coefficients. The normalization rule must be derived by a physical theory rather than selected solely for numerical convenience.

Within the conditional TCGS–SEQUENTION interpretation,  $\mathcal{P}_{\min}$  and  $\mathcal{P}_{-t}$  perform different logical work. Minimal smooth completion fixes four rather than five or more source dimensions and is a parsimony postulate. Exclusion of ontic time fixes the dimensional type of the additional direction. Neither follows from spectral flow, the preimage theorem, or the Schwarzschild calibration. The source signature and source-to-causality map remain decisive because a Lorentzian source risks reproducing ordinary spacetime, while a Riemannian source must generate Lorentzian propagation, null cones, and the invariant speed.

The scope of the central theorem is consequently precise. It proves compatibility between a three-dimensional carrier and a nonempty two-dimensional regular boundary. It does not discriminate among source ontologies. Its value lies in eliminating a category error: neither a two-dimensional threshold nor an ultraviolet spectral dimension near two entails topological collapse of the carrier. Positive evidential support for a source architecture must come from independently derived normalization, transversality, causal structure, selector transport, and novel observable consequences.

## 16 Conclusion

Planck units are exact dimensional constructions from  $\hbar$ ,  $c$ , and  $G$ . They do not by themselves determine a minimum-length ontology, a microscopic lattice, a running spectral dimension, or a topological reduction of the physical carrier.

A universal Planck length is not a universal rigid cell. A classical cube with rest edge  $\ell_P$  has boosted longitudinal edge  $\ell_P/\gamma_v$  and simultaneous volume  $\ell_P^3/\gamma_v$ . A regular lattice of such cells would therefore select a preferred rest frame under ordinary Lorentz transformations. Quantum spectra, causal sprinklings, deformed boosts, spectral flow, and scalar thresholds remain separate possibilities because their transformation laws and physical objects differ.

Several quantum-gravity programmes derive an ultraviolet spectral dimension near two. For a declared diffusion generator,

$$d_s(\sigma_D) = -2 \frac{d \log P(\sigma_D)}{d \log \sigma_D}, \quad (126)$$

and CDT, Hořava–Lifshitz gravity, and asymptotic-safety calculations produce flows from approximately four in the infrared to approximately two in the ultraviolet. These are analytic properties of specific operators and ensembles; they do not require topological dimension to change.

The scale-free source-interface construction is

$$\Omega_C = \{\mathcal{B}_C \leq 1\} \subset \mathcal{C}^4, \quad \Gamma = \mathcal{B}_C^{-1}(1), \quad (127)$$

$$X : \Sigma^3 \hookrightarrow \mathcal{C}^4, \quad \Omega_\Sigma = X^{-1}(\Omega_C), \quad \mathcal{H}_\Sigma = X^{-1}(\Gamma). \quad (128)$$

If  $X \pitchfork \Gamma$ , then  $\Omega_\Sigma$  is a three-dimensional manifold with boundary  $\partial\Omega_\Sigma = \mathcal{H}_\Sigma$ . The boundary may be empty; if nonempty, it is two-dimensional. If  $X(\Sigma) \subseteq \Gamma$ , the carrier is threshold-saturated and  $\mathcal{H}_\Sigma = \Sigma^3$ . Tangential contact can produce singular or variable-dimensional preimages. The theorem therefore supplies a precise classification rather than an unconditional two-dimensional threshold.

The theorem is intrinsically scale free. The labels  $\mathcal{B}_\Pi$ ,  $\Omega_\Pi$ , and  $\mathcal{H}_\Pi$  require a normalization bridge connecting  $\mathcal{B}_C$  to the observed Planck package. An intrinsic source normalization would have to yield  $\hbar$ ,  $c$ , and  $G$  under readout. A Shadow-imported normalization is a conditional calibration and depends on the causal and gravitational readout structure it uses. This distinction prevents the Planck label from being inserted into the source geometry by notation alone.

Covariance is equally structured. Pure relabelings make scalar naturality automatic. For a leaf-changing comparison with source lift  $\mathfrak{f}_\kappa$ ,

$$\mathcal{B}_\Sigma^{(\sigma')} \circ \varphi_\kappa - \mathcal{B}_\Sigma^{(\sigma)} = X_\sigma^*(\mathfrak{f}_\kappa^* \mathcal{B}_C - \mathcal{B}_C). \quad (129)$$

A lifted one-parameter boost family therefore requires  $\mathcal{L}_\xi \mathcal{B}_C = 0$  on the relevant source region. This gives a determinate field-classification condition, while the physical construction of the source lift remains part of the source-to-causality bridge. When the readout-calibrated route fixes  $b_0$  on a transverse reference leaf and the lifted flow sweeps a region once, the field is no longer underdetermined there: Proposition 5.6 gives the unique characteristic extension  $\mathcal{B}_C(\mathfrak{f}_\eta(y)) = b_0(y)$ .

The exponent two in  $\ell_P^2 \mathcal{K}$  compensates the inverse-square units of curvature. Black-hole entropy proportional to area expresses state counting on a horizon. Neither statement fixes bulk topology. The topological number two in a nonempty  $\mathcal{H}_\Sigma$  is the dimension of a codimension-one boundary; the spectral number two in  $d_s \rightarrow 2$  is a heat-trace exponent. Their equality is numerical until a physical operator theorem connects them.

The Schwarzschild calculation remains a separate Shadow-spacetime result. With

$$\hat{\mathcal{B}}_K = \ell_P^2 \sqrt{R_{\mu\nu\rho\sigma} [g_{\text{Schw}}] R^{\mu\nu\rho\sigma} [g_{\text{Schw}}]}, \quad (130)$$

its unit level intersects a spatial slice in

$$\mathcal{S}_K(t) \cong S_{r_K}^2, \quad r_K = (2\sqrt{3})^{1/3}(\ell_P^2 r_s)^{1/3}. \quad (131)$$

For a solar-mass black hole  $r_K \simeq 1.39 \times 10^{-22}$  m. The horizon coincidence occurs at  $M_\star \simeq 0.9306 m_P$  and  $r_{s,\star} \simeq 1.8612 \ell_P$ . These results do not identify  $\mathcal{S}_K(t)$  with  $\mathcal{H}_\Sigma$ ; the equality requires the source-to-effective-metric bridge in Eq. (93).

Within TCGS-SEQUENTION, minimal smooth completion fixes the source count at four only as a parsimony premise, and exclusion of ontic time classifies the additional direction only as a separate typing premise. The source signature, Lorentzian causal reconstruction, normalization package, physical immersion, transversality, selector groupoid, source comparison action, diffusion generator, and source-to-Schwarzschild map remain constructive requirements. The  $\mathcal{A}_\Pi$  substitution remains a rescaling ansatz until it produces a nontrivial observable consequence.

The central result is therefore a controlled compatibility statement. Ultraviolet propagation can be spectrally two-dimensional, a regular boundary can be topologically two-dimensional, a Schwarzschild curvature diagnostic can define a two-sphere, and the associated carrier can remain topologically three- or four-dimensional. A complete physical theory must identify which object carries each dimension, derive the bridges among those objects, and supply predictions that distinguish its source architecture from alternative realizations.

## A Full Dimensional Derivation of the Planck Generator

Starting from  $Q = \hbar^a c^b G^d$ , the dimensions are

$$[Q] = M^{a-d} L^{2a+b+3d} T^{-a-b-2d}. \quad (132)$$

Matching  $M^m L^\ell T^\tau$  gives Eq. (5). Solving yields Eq. (6), and hence Eq. (7). The standard examples are

$$Q_P(0, 1, 0) = \sqrt{\frac{\hbar G}{c^3}} = \ell_P, \quad (133)$$

$$Q_P(0, 0, 1) = \sqrt{\frac{\hbar G}{c^5}} = t_P, \quad (134)$$

$$Q_P(1, 0, 0) = \sqrt{\frac{\hbar c}{G}} = m_P, \quad (135)$$

$$Q_P(1, 2, -2) = \sqrt{\frac{\hbar c^5}{G}} = E_P, \quad (136)$$

$$Q_P(1, -3, 0) = \frac{c^5}{\hbar G^2} = \rho_P. \quad (137)$$

The generator is unique only up to dimensionless factors. Any claim about a precise threshold must therefore state the physical invariant and numerical convention that fix those factors.

## B Algebraic Consequences of the $\mathcal{A}_\Pi$ Ansatz

Substituting  $G\mathcal{A}_\Pi$  for  $G$  in Eq. (7) gives

$$Q_\Pi = Q_P \mathcal{A}_\Pi^d, \quad d = \frac{\ell + \tau - m}{2}. \quad (138)$$

Thus

$$\ell_\Pi = \ell_P \mathcal{A}_\Pi^{1/2}, \quad (139)$$

$$t_\Pi = t_P \mathcal{A}_\Pi^{1/2}, \quad (140)$$

$$m_\Pi = m_P \mathcal{A}_\Pi^{-1/2}, \quad (141)$$

$$E_\Pi = E_P \mathcal{A}_\Pi^{-1/2}, \quad (142)$$

$$\rho_\Pi = \rho_P \mathcal{A}_\Pi^{-2}. \quad (143)$$

Products whose net  $G$  exponent is zero automatically lose all  $\mathcal{A}_\Pi$  dependence. For example,

$$\ell_\Pi E_\Pi = \hbar c, \quad t_\Pi E_\Pi = \hbar, \quad m_\Pi \ell_\Pi = \frac{\hbar}{c}. \quad (144)$$

These are dimensional identities, not independent evidence that the ansatz captures projection geometry.

## C Notation and Type Dictionary

Table 5: Principal notation.

| Symbol                                       | Meaning                                                                                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{C}^4$                              | Smooth four-dimensional source representation used in the conditional TCGS-SEQUENTATION map; not asserted to exhaust ontic Territory.              |
| $\Sigma^3$                                   | Three-dimensional Shadow carrier.                                                                                                                  |
| $X : \Sigma^3 \hookrightarrow \mathcal{C}^4$ | Rank-three immersion or embedding selector.                                                                                                        |
| $X^*$                                        | Pullback of declared source fields or tensors to the carrier.                                                                                      |
| $\text{Ctd}_{4,S,G}^\sigma$                  | Typed constitutive source-to-Shadow readout relation.                                                                                              |
| $\mathcal{J}_C$                              | Declared source invariant package used, if available, to construct an admissibility field.                                                         |
| $\mathfrak{P}$                               | Normalization data entering $\mathfrak{N}_C[\mathcal{J}_C; \mathfrak{P}]$ ; source-derived or explicitly imported from a reference Shadow readout. |
| $\mathcal{B}_C$                              | Smooth dimensionless source admissibility field; not Planck normalized by notation alone.                                                          |
| $\mathcal{L}_0, b_0$                         | Embedded reference calibration leaf and its readout-calibrated scalar data in Eqs. (26)–(28).                                                      |
| $\xi, \mathfrak{f}_\eta$                     | Generator and local flow of a leaf-changing source comparison action.                                                                              |
| $U_{\text{sw}}$                              | Source region swept once by the comparison flow in Proposition 5.6.                                                                                |
| $\Omega_C$                                   | Source domain $\{\mathcal{B}_C \leq 1\}$ .                                                                                                         |
| $\Gamma$                                     | Regular source interface $\mathcal{B}_C^{-1}(1) = \partial\Omega_C$ .                                                                              |
| $\mathcal{B}_\Sigma$                         | Carrier scalar $X^*\mathcal{B}_C = \mathcal{B}_C \circ X$ .                                                                                        |

*Continued on the next page*

Table 5 continued

| Symbol                                         | Meaning                                                                                                                     |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $\Omega_\Sigma$                                | Carrier preimage $X^{-1}(\Omega_C)$ ; a three-manifold with boundary under Theorem 4.2.                                     |
| $\mathcal{H}_\Sigma$                           | Boundary $X^{-1}(\Gamma)$ ; possibly empty, and two-dimensional when nonempty and transverse.                               |
| $\mathcal{B}_\Pi, \Omega_\Pi, \mathcal{H}_\Pi$ | Planck-labelled versions of $\mathcal{B}_\Sigma, \Omega_\Sigma, \mathcal{H}_\Sigma$ , permitted only after Definition 4.1.  |
| $d_{\text{top}}$                               | Topological dimension of a carrier or submanifold.                                                                          |
| $d_s(\sigma_D)$                                | Scale-dependent spectral dimension defined from a declared return probability.                                              |
| $\sigma_D$                                     | Diffusion scale; not physical time or the selector symbol $\sigma$ .                                                        |
| $\text{Sel}_\Xi$                               | Threshold-independent selector-equivalence groupoid for a fixed source configuration.                                       |
| $\varphi_\kappa$                               | Carrier comparison diffeomorphism assigned to an independently established selector arrow.                                  |
| $\mathfrak{f}_\kappa$                          | Source lift of a leaf-changing selector comparison, when it exists.                                                         |
| $\mathcal{D}$                                  | Positive diffusion generator whose heat kernel defines $d_s$ .                                                              |
| $\mathcal{N}_P[\mathcal{K}]$                   | Shadow-side dimensionless curvature normalization $\ell_P^2 \mathcal{K}$ ; not automatically a source pullback.             |
| $\hat{\mathcal{B}}_K$                          | Independent Schwarzschild Shadow-spacetime curvature diagnostic.                                                            |
| $\mathcal{S}_K(t)$                             | Spatial sphere $i_t^{-1}(\hat{\mathcal{B}}_K^{-1}(1))$ in the Schwarzschild map; not identified with $\mathcal{H}_\Sigma$ . |
| $r_K$                                          | Radius of $\mathcal{S}_K(t)$ , defined by $\hat{\mathcal{B}}_K(r; M) = 1$ .                                                 |
| $G$                                            | Source geometric structure or metric representation; signature open.                                                        |
| $g_{\mu\nu}^{\text{eff}}$                      | Effective Lorentzian Shadow spacetime metric, whose source derivation remains open.                                         |
| $\mathcal{A}_\Pi$                              | Exploratory dimensionless rescaling ansatz, not an established invariant or field.                                          |

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